FRM PART II • 25 NOVEMBER 2026

Master LO Map

LO-level priority • Vardeez + Recall Workbooks

All 107 readings, all 591 LOs, in syllabus order. Chapter names and LO wording verbatim from the official GARP 2026 FRM Part II Learning Objectives booklet.

Colour is set per LO, by the concept it carries, not by the chapter.

P1	A Tier A workbook topic (August recall core) hits this LO's concept.	143 LOs
P2	A Tier B topic hits it, or Vardeez flags it inside a Very Important chapter.	112 LOs
P3	A Tier C or D topic hits it, or Vardeez flags it in any other chapter.	130 LOs
P4	Neither source flags it. Still examinable.	206 LOs

Under each LO

WORKBOOK A6 + VARDEEZ FOCUS	Why the LO got its band.
►	Vardeez's specific pointer inside the LO.
A6·S3	A reported exam shape from your workbook whose concept sits in this LO (section, shape number).
×	Vardeez says skip. A Tier A hit still makes it P1.

Each shape was matched to the LO whose official wording covers it, even where that falls outside the LO range the workbook listed (e.g. A15 hazard-rate shapes land on CR-9 d, e, h). A shape that spans two concepts is shown under both LOs. Workbook chips: A B C D. Sections B5 (climate) and D17 (2023 bank failures) are not in the 2026 LOs and are left out.

P1 checklist

143 LOs

MARKET RISK MEASUREMENT AND MANAGEMENT

☐	MR-1 b	Estimate VaR using a parametric approach for both normal and lognormal asset price distributions.	A13
☐	MR-2 a	Apply the bootstrap historical simulation approach to estimate coherent risk measures.	A19
☐	MR-2 c	Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches.	A19
☐	MR-2 d	Identify advantages and disadvantages of non-parametric estimation methods.	A19
☐	MR-3 b	Describe extreme value theory (EVT) and its use in risk management.	A17
☐	MR-3 c	Describe the peaks-over-threshold (POT) approach.	A17
☐	MR-3 d	Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks.	A17
☐	MR-3 e	Discuss the application of the generalized Pareto (GP) distribution in the POT approach.	A17
☐	MR-8 c	Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.	A8
☐	MR-8 d	Describe the structure, uses, and payoffs of a correlation swap.	A8
☐	MR-8 g	Explain how correlation risk relates to systemic and concentration risk.	A8
☐	MR-14 a	Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift.	A7
☐	MR-14 b	Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift.	A7
☐	MR-14 d	Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift.	A7
☐	MR-14 e	Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices.	A7
☐	MR-14 f	Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion.	A7
☐	MR-14 g	Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life.	A25
☐	MR-15 a	Describe the short-term rate process under a model with time-dependent volatility.	A7
☐	MR-15 b	Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility.	A7
☐	MR-16 a	Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics.	A25
☐	MR-16 b	Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model.	A25
☐	MR-16 c	Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model.	A25
☐	MR-17 a	Describe a volatility smile and volatility skew.	A3
☐	MR-17 b	Explain the implications of put-call parity on the implied volatility of call and put options.	A3
☐	MR-17 c	Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset.	A3
☐	MR-17 d	Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility.	A3
☐	MR-17 e	Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape.	A3
☐	MR-18 a	Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes.	A4
☐	MR-18 b	Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons.	A4
☐	MR-18 c	Explain the FRTB revisions to Basel regulations in the following areas: - Classification of positions in the trading book compared to the banking book. - Backtesting, profit and loss attribution, credit risk, and securitizations.	A4

CREDIT RISK MEASUREMENT AND MANAGEMENT

☐	CR-2 a	Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control.	A22
☐	CR-4 d	Define and calculate unexpected loss (UL).	A1
☐	CR-4 e	Estimate the variance of default probability assuming a binomial distribution.	A1

☐	**CR-4 f**	Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size.	A1
☐	**CR-4 g**	Describe how economic capital is derived.	A1
☐	**CR-5 d**	Describe the judgmental approaches, empirical models, and financial models to predict default.	A10
☐	**CR-5 e**	Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model.	A5
☐	**CR-5 g**	Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan.	A2
☐	**CR-9 b**	Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality.	A10
☐	**CR-9 d**	Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default.	A15
☐	**CR-9 e**	Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities.	A5, A15
☐	**CR-9 h**	Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates.	A15
☐	**CR-9 k**	Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk.	A5
☐	**CR-9 l**	Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model.	A5
☐	**CR-9 m**	Calculate distance to default and default probability using the Merton model.	A5
☐	**CR-10 d**	Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings- based (IRB) approach.	A31
☐	**CR-16 d**	Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages.	A21
☐	**CR-17 b**	Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount.	A21
☐	**CR-17 c**	Calculate the credit support amount (margin) under various scenarios.	A21
☐	**CR-17 f**	Explain the process for the reconciliation of collateral disputes.	A21
☐	**CR-17 g**	Explain the features of a collateralization agreement.	A21
☐	**CR-17 h**	Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality.	A21
☐	**CR-17 j**	Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization.	A21
☐	**CR-19 h**	Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount.	A21
☐	**CR-20 c**	Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization.	A6
☐	**CR-20 e**	Describe debt value adjustment (DVA) and bilateral CVA (BCVA).	A6
☐	**CR-20 f**	Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA.	A6
☐	**CR-20 g**	Calculate DVA, BCVA, and BCVA as a spread.	A6
☐	**CR-20 i**	Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread.	A6
☐	**CR-20 j**	Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins.	A21
☐	**CR-20 k**	Describe wrong-way risk and contrast it with right-way risk.	A20
☐	**CR-20 l**	Identify examples of wrong-way risk and examples of right-way risk.	A20
☐	**CR-20 m**	Discuss the impact of collateral on wrong-way risk.	A20
☐	**CR-20 n**	Identify examples of wrong-way collateral.	A20
☐	**CR-20 o**	Discuss the impact of wrong-way risk on central counterparties (CCPs).	A20
☐	**CR-20 p**	Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches.	A20
☐	**CR-22 c**	Describe a waterfall structure in a securitization.	A9
☐	**CR-22 e**	Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.	A9
☐	**CR-22 f**	Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year.	A9
☐	**CR-23 g**	Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures.	A9
☐	**CR-23 h**	Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate.	A9

OPERATIONAL RISK AND RESILIENCE

☐	**ORR-1 b**	Describe the seven Basel II event risk categories and identify examples of operational risk events in each category.	A22
☐	**ORR-1 d**	Describe operational resilience, identify the elements of an operational resilience framework, and summarize regulatory expectations for operational resilience.	A23
☐	**ORR-2 b**	Describe and compare the roles of different committees and the board of directors in operational risk governance.	A22
☐	**ORR-2 c**	Describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense.	A22
☐	**ORR-2 d**	Explain best practices and regulatory expectations for the development of a risk appetite for operational risk and for a strong risk culture.	A22
☐	**ORR-3 d**	Describe and apply the Level 1, 2, and 3 categories in the Basel operational risk taxonomy.	A22
☐	**ORR-4 f**	Describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations.	A10, A17
☐	**ORR-4 g**	Describe and explain the steps to ensure a strong level of operational resilience, and to test the operational resilience of important business services.	A23
☐	**ORR-8 b**	Describe current practices by banks and supervisors in the governance of a cyber-risk management framework, including roles and responsibilities.	A23
☐	**ORR-8 c**	Explain methods for supervising cyber-resilience, testing and incident response approaches, and cybersecurity and resilience metrics.	A23
☐	**ORR-9 a**	Provide examples of cyber threats and information security risks, and describe frameworks and best practices for managing cyber risks.	A23
☐	**ORR-9 b**	Describe lessons learned from the Equifax case study.	A23
☐	**ORR-10 a**	Explain best practices recommended by the Basel committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks.	A29
☐	**ORR-10 b**	Describe recommended practices for the acceptance, verification, and identification of customers at a bank.	A29
☐	**ORR-10 c**	Explain practices for managing ML/FT risks in a group-wide and cross-border context.	A29
☐	**ORR-11 a**	Describe elements of a control framework to manage financial fraud risk and money laundering risk.	A29
☐	**ORR-18 c**	Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance.	A2
☐	**ORR-18 e**	Calculate the hurdle rate and apply this rate in making business decisions using RAROC.	A2
☐	**ORR-18 f**	Calculate the adjusted RAROC for a project to determine its viability.	A2
☐	**ORR-21 c**	Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book.	A4
☐	**ORR-21 e**	Compare the standardized internal ratings-based (IRB) approach, the foundation IRB approach, and the advanced IRB approach for the calculation of credit risk capital under Basel II.	A31
☐	**ORR-21 f**	Calculate credit risk capital under Basel II utilizing the IRB approach.	A31
☐	**ORR-22 a**	Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge.	A4
☐	**ORR-22 f**	Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio.	A32
☐	**ORR-23 b**	Summarize the December 2017 revisions to the Basel III framework in the following areas: - The standardized approach to credit risk - The internal ratings-based (IRB) approaches for credit risk - The CVA risk framework - The operational risk framework - The leverage ratio framework	A31
☐	**ORR-23 c**	Describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor.	A31

LIQUIDITY AND TREASURY RISK MEASUREMENT AND MANAGEMENT

☐	**LTR-1 a**	Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).	A11
☐	**LTR-2 f**	Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.	A11
☐	**LTR-4 a**	Compare various money market and capital market instruments and discuss their advantages and disadvantages.	A30
☐	**LTR-4 b**	Identify and discuss various factors that affect the choice of investment securities by a bank.	A30
☐	**LTR-5 a**	Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.	A30
☐	**LTR-5 b**	Compare strategies that a bank can use to meet demands for additional liquidity.	A30

☐	**LTR-5 c**	Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).	A26
☐	**LTR-7 d**	Interpret the term structure of expected cash flows and cumulative cash flows.	A26
☐	**LTR-7 e**	Discuss the impact of available asset transactions on cash flows and liquidity generation capacity.	A26
☐	**LTR-12 a**	Distinguish between the various sources of non-deposit liabilities at a bank.	A27
☐	**LTR-12 c**	Discuss factors affecting the choice of non-deposit funding sources.	A27
☐	**LTR-12 d**	Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.	A27
☐	**LTR-14 a**	Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.	A27
☐	**LTR-14 c**	Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost).	A27
☐	**LTR-17 b**	Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.	A16
☐	**LTR-17 c**	Describe duration gap management and apply this strategy to protect a bank's net worth.	A16

RISK MANAGEMENT AND INVESTMENT MANAGEMENT

☐	**IM-3 b**	Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.	A18
☐	**IM-3 d**	Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law.	A18
☐	**IM-5 a**	Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.	A12
☐	**IM-5 c**	Apply the concept of marginal VaR in making portfolio management decisions.	A12, A18
☐	**IM-5 d**	Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk.	A12
☐	**IM-7 b**	Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant.	A18
☐	**IM-7 c**	Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.	A18
☐	**IM-7 i**	Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.	A18
☐	**IM-8 a**	Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles.	A24
☐	**IM-8 b**	Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies.	A24
☐	**IM-8 c**	Describe the mechanics and risks of event-driven strategies such as merger arbitrage.	A24
☐	**IM-8 d**	Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy.	A24
☐	**IM-9 a**	Describe different risks faced by investors in hedge funds.	A24
☐	**IM-9 b**	Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk.	A24
☐	**IM-9 c**	Summarize different approaches taken by regulators to mitigate risks related to hedge funds.	A24
☐	**IM-9 d**	Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure.	A24
☐	**IM-10 c**	Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability.	A28
☐	**IM-10 d**	Assess potential policy recommendations that could help mitigate the risks associated with private credit.	A28
☐	**IM-11 b**	Describe different investment structures that can be used to invest in private assets and compare their characteristics.	A28
☐	**IM-11 c**	Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages.	A28
☐	**IM-11 e**	Describe the practice of co-investing in private markets and assess its advantages and disadvantages.	A28
☐	**IM-11 f**	Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund.	A28
☐	**IM-14 b**	Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk.	A24
☐	**IM-17 c**	Assess the impact of biases on reported returns for illiquid assets.	A24

CURRENT ISSUES IN FINANCIAL MARKETS

☐	**CI-1 a**	Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI.	A14
☐	**CI-1 c**	Explain the implications of further adoption of AI on financial stability.	A14
☐	**CI-2 a**	Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers.	A14
☐	**CI-2 c**	Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: - Third-party dependencies and service provider concentration - Market correlations - Cyber risks - Model risk, data, and governance - Other vulnerabilities, such as fraud, disinformation, and misalignment	A14
☐	**CI-3 b**	Discuss the potential drivers of private credit growth and interpret regression results about their impact.	A28
☐	**CI-3 c**	Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them.	A28

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ID	Reading	P1 LOs	P2 LOs	P3 LOs
MARKET RISK MEASUREMENT AND MANAGEMENT • 20% • 16 Qs				
MR-1	Estimating Market Risk Measures: An Introduction and Overview	b	c, e, f	–
MR-2	Non-Parametric Approaches	a, c, d	–	–
MR-3	Parametric Approaches (II): Extreme Value	b, c, d, e	–	–
MR-4	Backtesting VaR	–	–	c, d, e, f
MR-5	VaR Mapping	–	–	c, d, f
MR-6	Validating Bank Holding Companies' Value-at-Risk Models for Market Risk	–	–	a, b, c, d
MR-7	Beyond Exceedance-Based Backtesting of Value-at-Risk Models	–	–	a, b, c, d
MR-8	Correlation Basics: Definitions, Applications, and Terminology	c, d, g	–	–
MR-9	Empirical Properties of Correlation: How Do Correlations Behave in the Real World?	–	–	b, c
MR-10	Financial Correlation Modeling – Bottom-Up Approaches	–	b	–
MR-11	Regression Hedging and Principal Component Analysis	–	–	a, b, c, d, h
MR-12	Arbitrage Pricing with Term Structure Models	–	–	d, f, h
MR-13	Expectations, Risk Premium, Convexity, and the Shape of the Term Structure	–	–	a, b, c, d
MR-14	The Art of Term Structure Models: Drift	a, b, d, e, f, g	–	c
MR-15	The Art of Term Structure Models: Volatility and Distribution	a, b	d, e, f	–
MR-16	The Vasicek and Gauss+ Models	a, b, c	d	–
MR-17	Volatility Smiles	a, b, c, d, e	f, g, h, i	–
MR-18	Fundamental Review of the Trading Book	a, b, c	–	–
CREDIT RISK MEASUREMENT AND MANAGEMENT • 20% • 16 Qs				
CR-1	Fundamentals of Credit Risk	–	–	–
CR-2	Governance	a	–	c
CR-3	Credit Risk Management	–	–	e, g, h
CR-4	Capital Structure in Banks	d, e, f, g	c	–
CR-5	Introduction to Credit Risk Modeling and Assessment	d, e, g	f	c
CR-6	Credit Scoring and Rating	–	–	b, d
CR-7	Credit Scoring and Retail Credit Risk Management	–	–	b, c, f, h
CR-8	Country Risk: Determinants, Measures, and Implications	–	–	g
CR-9	Estimating Default Probabilities	b, d, e, h, k, l, m	a, c, f, j, n	g
CR-10	Credit Value at Risk	d	b, e, f	–

ID	Reading	P1 LOs	P2 LOs	P3 LOs
CR-11	Portfolio Credit Risk	–	a, c, d, f, h	–
CR-12	Credit Risk	–	f	c, e, g
CR-13	Credit Derivatives	–	–	a, b, f, g
CR-14	Derivatives	–	–	i
CR-15	Counterparty Risk and Beyond	–	–	a, b, c, d, e, f, g, h
CR-16	Netting, Close-out and Related Aspects	d	e, f	–
CR-17	Margin (Collateral) and Settlement	b, c, f, g, h, j	–	–
CR-18	Central Clearing	–	–	f
CR-19	Future Value and Exposure	h	d, e, f, g	a
CR-20	CVA	c, e, f, g, i, j, k, l, m, n, o, p	a, b, d, h, q	–
CR-21	The Evolution of Stress Testing Counterparty Exposures	–	a, b, c, d, e, f, g, h	–
CR-22	Structured Credit Risk	c, e, f	i, l	h
CR-23	An Introduction to Securitisation	g, h	–	c
OPERATIONAL RISK AND RESILIENCE • 20% • 16 Qs				
ORR-1	Introduction to Operational Risk and Resilience	b, d	–	–
ORR-2	Risk Governance	b, c, d	–	–
ORR-3	Risk Identification	d	–	–
ORR-4	Risk Measurement and Assessment	f, g	–	a, b, c
ORR-5	Risk Mitigation	–	–	b, c
ORR-6	Risk Reporting	–	–	–
ORR-7	Integrated Risk Management	–	–	b
ORR-8	Cyber-resilience: Range of Practices	b, c	–	e
ORR-9	Cyberthreats and Information Security Risks	a, b	–	–
ORR-10	Sound Management of Risks related to Money Laundering and Financing of Terrorism	a, b, c	–	–
ORR-11	Financial Crime and Fraud	a	b	–
ORR-12	Guidance on Managing Outsourcing Risk	–	a, b, c	–
ORR-13	Third-Party Risk Management	–	a, b	–
ORR-14	Investor Protection and Compliance Risks in Investment Activities	–	a, b	–
ORR-15	Supervisory Guidance on Model Risk Management	–	a, b, c, d	–
ORR-16	Case Study: Model Risk and Model Validation	–	a, b, c	–

ID	Reading	P1 LOs	P2 LOs	P3 LOs
ORR-17	Stress Testing Banks	–	–	a, b
ORR-18	Risk Capital Attribution and Risk-Adjusted Performance Measurement	c, e, f	–	a
ORR-19	Range of Practices and Issues in Economic Capital Frameworks	–	–	–
ORR-20	Capital Planning at Large Bank Holding Companies	–	–	a, b
ORR-21	Capital Regulation Before the Global Financial Crisis	c, e, f	–	a, b, d
ORR-22	Solvency, Liquidity and Other Regulation After the Global Financial Crisis	a, f	–	b, c, d, e, g
ORR-23	High-level Summary of Basel III Reforms	b, c	a	–
ORR-24	Basel III: Finalising Post-Crisis Reforms	–	–	b, c
LIQUIDITY AND TREASURY RISK MEASUREMENT AND MANAGEMENT • 15% • 12 Qs				
LTR-1	Liquidity Risk	a	–	b, d
LTR-2	Liquidity and Leverage	f	–	–
LTR-3	Early Warning Indicators	–	a, b, c	–
LTR-4	The Investment Function in Financial-Services Management	a, b	c	–
LTR-5	Liquidity and Reserves Management: Strategies and Policies	a, b, c	–	–
LTR-6	Intraday Liquidity Risk Management	–	–	a, c
LTR-7	Monitoring Liquidity	d, e	–	a, b
LTR-8	The Failure Mechanics of Dealer Banks	–	–	a, b, c
LTR-9	Liquidity Stress Testing	–	–	b
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LTR-12	Managing Non-Deposit Liabilities	a, c, d	–	b
LTR-13	Repurchase Agreements and Financing	–	e, f	a, c
LTR-14	Liquidity Transfer Pricing: A Guide to Better Practice	a, c	–	–
LTR-15	The US Dollar Shortage in Global Banking and the International Policy Response	–	a, b, c	–
LTR-16	Covered Interest Parity Lost: Understanding the Cross-Currency Basis	–	–	a, b, c
LTR-17	Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	b, c	a, d	–
RISK MANAGEMENT AND INVESTMENT MANAGEMENT • 15% • 12 Qs				
IM-1	Factor Theory	–	–	b, c, d
IM-2	Factors	–	d, e	c
IM-3	Alpha (and the Low-Risk Anomaly)	b, d	a, e, f, h, i	–
IM-4	Portfolio Construction	–	d, g, h	b, c
IM-5	Portfolio Risk: Analytical Methods	a, c, d	–	–

ID	Reading	P1 LOs	P2 LOs	P3 LOs
IM-6	VaR and Risk Budgeting in Investment Management	–	e, h	–
IM-7	Portfolio Performance Evaluation	b, c, i	a, e	–
IM-8	Hedge Fund Investment Strategies	a, b, c, d	–	–
IM-9	Risk, Regulation and Organizational Structure	a, b, c, d	–	–
IM-10	The Rise and Risks of Private Credit	c, d	–	–
IM-11	Private Markets Investing	b, c, e, f	–	–
IM-12	Performing Due Diligence on Specific Managers and Funds	–	–	c, e
IM-13	Private Capital Investing: Distress Symptoms and Remedies	–	–	a, b
IM-14	Madoff: A Riot of Red Flags	b	a	–
IM-15	Market-Driven Scenarios: An Approach for Plausible Scenario Construction	–	b, c	a
IM-16	Liquidity Risk Management	–	a, b, c, d, e	–
IM-17	Illiquid Assets	c	e, f	d
CURRENT ISSUES IN FINANCIAL MARKETS • 10% • 8 Qs				
CI-1	Advances in Artificial Intelligence: Implications for Capital Markets Activities	a, c	b, d	–
CI-2	The Financial Stability Implications of Artificial Intelligence	a, c	b	–
CI-3	The Global Drivers of Private Credit	b, c	a	–
CI-4	Global Financial Stability Report	–	–	a, b, c, d, e, f
CI-5	Monetary and fiscal policy: safeguarding stability and trust	–	–	a, b, c
CI-6	Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets	–	–	a, b, c, d
CI-7	Tokenization and Financial Market Inefficiencies	–	a, b, c, d, e	–
CI-8	Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector	–	–	a, b, c

Market Risk Measurement and Management

20% • 16 Qs

MR-1 Estimating Market Risk Measures: An Introduction and Overview

P1 ×1 P2 ×3 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *a) calc of Normal/Lognormal VaR b) calc of ES c) Interpretation of standard Errors d) Interpretation of QQ plots*

RECALL WORKBOOK SECTIONS

A13 Lognormal VaR

Tier A • Aug ✓ • Map #14 (6/16) • TG ✓ • 4 Qs • Vol 1 p.101 • LOs a to f

B2 Expected Shortfall and Coherence

Tier B • Map #12 (6/16) • TG ✓ • 3 Qs • Vol 2 p.19 • LOs e

P4 a Estimate VaR using a historical simulation approach.

P1 **b Estimate VaR using a parametric approach for both normal and lognormal asset price distributions.**

WORKBOOK A13 + VARDEEZ FOCUS

▶ Vardeez: Calculation of normal and lognormal VaR

A13·S1 choose between two volatilities, 16% rising to 18% six weeks ago: which regime governs the VaR horizon

A13·S2 direct lognormal VaR from the formula, the plain form seen in August

A13·S3 lognormal form $1 - e^{\mu - z\sigma}$ kept distinct from the normal arithmetic form $-\mu + \sigma z$

P2 **c Estimate the expected shortfall given profit and loss (P&L) or return data.**

WORKBOOK B2 + VARDEEZ FOCUS

▶ Vardeez: Calculation of ES

B2·S1 ES from a sample and its bootstrap resamples: exactly which losses enter the tail average, 4, 5 or 6

B2·S2 ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces under FRTB

P4 d Estimate risk measures by estimating quantiles.

P2 **e Evaluate estimators of risk measures by estimating their standard errors.**

VARDEEZ FOCUS

▶ Vardeez: Interpretation of standard errors

P2 **f Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution.**

VARDEEZ FOCUS

▶ Vardeez: Interpretation of QQ plots

MR-2 Non-Parametric Approaches

P1 ×3 P4 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *a) know abt Bootstrapping method HS method b) Volatility weighted method c) Filtered HS method*

RECALL WORKBOOK SECTIONS

A13 Lognormal VaR

Tier A • Aug ✓ • Map #14 (6/16) • TG ✓ • 4 Qs • Vol 1 p.101 • LOs c

A19 Non-Parametric VaR: Weighted HS, Bootstrap and FHS

Tier A • Aug ✓ • Map #28 (4/16) • TG ✓ • 5 Qs • Vol 1 p.140 • LOs a to d

B2 Expected Shortfall and Coherence

Tier B • Map #12 (6/16) • TG ✓ • 3 Qs • Vol 2 p.19 • LOs a, c, d

P1 **a** **Apply the bootstrap historical simulation approach to estimate coherent risk measures.**

WORKBOOK A19, B2 + VARDEEZ FOCUS

▶ Vardeez: Bootstrapping HS method

A19·S3 bootstrap HS property question

B2·S1 ES from a sample and its bootstrap resamples: exactly which losses enter the tail average, 4, 5 or 6

B2·S4 coherence: ES is subadditive, VaR is not (Q3)

P4 b Describe historical simulation using non-parametric density estimation.

P1 **c** **Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches.**

WORKBOOK A19, B2 + VARDEEZ FOCUS

▶ Vardeez: Volatility-weighted and filtered HS methods

A19·S1 age-weighted HS (BRW): single weights, cumulative weights and the weighted VaR read-off

A19·S2 volatility-weighted HS (HW), with the polarity trap: returns scaled up when current volatility is above the historical level

A19·S4 differences among the non-parametric approaches

A19·S5 expected shortfall in filtered historical simulation

B2·S3 ES within filtered historical simulation

P1 **d** **Identify advantages and disadvantages of non-parametric estimation methods.**

WORKBOOK A19

A19·S4 differences among the non-parametric approaches

MR-3 Parametric Approaches (II): Extreme Value

P1 ×4 P4 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *a) Describe EVT & its 3 general cases of GEV distribution b) Compare GEV & POT approach Note: Don't get too much into calculations here*

! Vardeez: do not go too deep into calculations here.

RECALL WORKBOOK SECTIONS

A17 Extreme Value Theory: GEV vs POT

Tier A • Aug ✓ • Map #25 (5/16) • TG ✓ • 2 Qs • Vol 1 p.127 • LOs a to f

P4 a Explain the importance and challenges of extreme values in risk management.

P1 **b** **Describe extreme value theory (EVT) and its use in risk management.**

WORKBOOK A17 + VARDEEZ FOCUS

▶ Vardeez: EVT and the 3 general cases of the GEV distribution

A17·S1 GEV theory: Fisher-Tippett, three parameters, the Frechet, Gumbel and Weibull cases

A17·S5 shape parameter $\xi > 0$ means fat tails

A17·S6 operational-risk framing of EVT and block maxima

P1 **c** **Describe the peaks-over-threshold (POT) approach.**

WORKBOOK A17

A17·S3 peaks over threshold: the GPBdH theorem and the GP distribution of excess losses

A17·S4 POT threshold set too low: more data, more bias, less extreme

P1 **d** **Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks.**

WORKBOOK A17 + VARDEEZ FOCUS

▶ Vardeez: Compare GEV vs POT

A17·S2 GEV versus POT: block maxima versus exceedances, data use, threshold

P1 **e** **Discuss the application of the generalized Pareto (GP) distribution in the POT approach.**

WORKBOOK A17

A17·S3 peaks over threshold: the GPBdH theorem and the GP distribution of excess losses

P4 f Explain the multivariate EVT for risk management.

MR-4 Backtesting VaR

P3 ×4 P4 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *a) Understand calculation of Failure Rates & Model Validation calculation (using 2 table) b) Basel penalty zones Note: Pls refer Crash course Notes for this chapter*

! Vardeez: refer to crash course notes for this chapter.

RECALL WORKBOOK SECTIONS

C2 Backtesting VaR: Exceedance vs PIT

Tier C • Map #21 (4/16) • TG ✓ • 3 Qs • Vol 2 p.41 • LOs a to f

P4	a	Describe backtesting and exceptions and explain the importance of backtesting VaR models.
P4	b	Explain the significant difficulties in backtesting a VaR model.
P3	**c**	Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. WORKBOOK C2 + VARDEEZ FOCUS ► Vardeez: Failure rates and model verification calculation (using the 2 tables) C2·S1 backtesting VaR: counting exceptions against the model C2·S3 Kupiec POF (unconditional coverage) versus Christoffersen (independence, conditional coverage)
P3	**d**	Identify and describe Type I and Type II errors in the context of a backtesting process. WORKBOOK C2 C2·S4 Type I versus Type II errors
P3	**e**	Explain the need to consider conditional coverage in the backtesting framework. WORKBOOK C2 C2·S3 Kupiec POF (unconditional coverage) versus Christoffersen (independence, conditional coverage)
P3	**f**	Describe the Basel rules for backtesting. WORKBOOK C2 + VARDEEZ FOCUS ► Vardeez: Basel penalty zones C2·S5 Basel penalty zones and the multiplier they set (Q1)

MR-5 VaR Mapping

P3 ×3 P4 ×4 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *a) Differentiate Principal Duration & Cashflow mapping b) calc of Principal & Duration mapping c) LO 5.f especially interpretation of Variance improvement Note: DO NOT WORRY abt Non Linear & Linear Derivatives (Huge calculations may not show up)*

RECALL WORKBOOK SECTIONS

C3 VaR Mapping

Tier C • Map #22 (4/16) • TG ✓ • 2 Qs • Vol 2 p.46 • LOs a to g

P4	a	Explain the principles underlying VaR mapping and describe the mapping process.
P4	b	Explain how the mapping process captures general and specific risks, and calculate these risks in a portfolio given a set of primitive risk factors.
P3	**c**	Differentiate among the three methods for mapping portfolios of fixed-income securities. WORKBOOK C3 + VARDEEZ FOCUS ► Vardeez: Principal vs duration vs cash-flow mapping C3·S1 difference among principal, duration and cash-flow mapping C3·S3 ranking the three methods by how faithfully they capture risk (Q2)
P3	**d**	Summarize how to map a fixed-income portfolio into positions of standard instruments. VARDEEZ FOCUS ► Vardeez: Calculation of principal and duration mapping
P4	e	Describe how mapping of risk factors can support stress testing.
P3	**f**	Explain how VaR can be calculated and used relative to a performance benchmark. WORKBOOK C3 + VARDEEZ FOCUS ► Vardeez: Interpretation of variance improvement (his "LO 5.f") C3·S2 VaR relative to a benchmark: tracking error VaR and the interpretation of variance improvement
P4	g	Describe the method of mapping forwards, forward rate agreements, interest rate swaps, and options. × Vardeez says skip: Do not worry about linear and non-linear derivatives mapping; heavy calculations unlikely

MR-6 Validating Bank Holding Companies' Value-at-Risk Models for Market Risk

P3 ×4 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *New Topic - Highly Testable*

! New topic, highly testable.

RECALL WORKBOOK SECTIONS

D10 BHC Capital Planning and VaR Model Validation

Tier D • Map #53 (2/16) • TG ✓ • 2 Qs • Vol 2 p.119 • LOs a to d

P3 **a** Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process.

WORKBOOK D10 + VARDEEZ FOCUS

D10·S3 validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2)

P3 **b** Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis.

WORKBOOK D10 + VARDEEZ FOCUS

D10·S3 validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2)

P3 **c** Describe the challenges a financial institution could face when calculating confidence intervals for VaR.

WORKBOOK D10 + VARDEEZ FOCUS

D10·S3 validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2)

P3 **d** Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them.

WORKBOOK D10 + VARDEEZ FOCUS

D10·S3 validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2)

MR-7 Beyond Exceedance-Based Backtesting of Value-at-Risk Models

P3 ×4 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *New Topic (Remember Backtesting is imp) Highly Testable*

! New topic, highly testable. Backtesting is important.

RECALL WORKBOOK SECTIONS

C2 Backtesting VaR: Exceedance vs PIT

Tier C • Map #21 (4/16) • TG ✓ • 3 Qs • Vol 2 p.41 • LOs a to d

P3 **a** Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest.

WORKBOOK C2 + VARDEEZ FOCUS

C2·S2 PIT-based versus exceedance-based backtesting

P3 **b** Explain how to derive probability integral transforms (PITs) in the context of validating a VaR model.

VARDEEZ FOCUS

P3 **c** Describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model.

VARDEEZ FOCUS

P3 **d** Describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test.

WORKBOOK C2 + VARDEEZ FOCUS

C2·S2 PIT-based versus exceedance-based backtesting

MR-8 Correlation Basics: Definitions, Applications, and Terminology

P1 ×3 P4 ×4 VARDEEZ: MODERATE WORKBOOK BEST: TIER A

Vardeez: *a) Quanto Option & Correlation Swap b) Concentration Ratio Concept & Calculation as well*

RECALL WORKBOOK SECTIONS

A8 Correlation Swaps, Quanto Options and Concentration Ratio

Tier A • Aug ✓ • Map #8 (7/16) • TG ✓ • 5 Qs • Vol 1 p.65 • LOs a to g

P4	a	Describe financial correlation risk and the areas in which it appears in finance.
P4	b	Explain how correlation contributed to the global financial crisis of 2007-2009.
P1	**c**	**Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.**

WORKBOOK A8 + VARDEEZ FOCUS

▶ Vardeez: Quanto options

A8·S4 quanto options: what correlation risk they carry and in which direction it moves the price

P1	**d**	**Describe the structure, uses, and payoffs of a correlation swap.**

WORKBOOK A8 + VARDEEZ FOCUS

▶ Vardeez: Correlation swap

A8·S1 correlation swap payoff to the fixed-rate payer, plain numerical as seen in August

A8·S2 the same swap from the seller's (fixed-rate receiver's) side, with irrelevant volatilities

A8·S3 correlation swap return per unit notional, reverse-solved for the strike

P4	e	Estimate the impact of different correlations between assets in the trading book on the VaR capital charge.
P4	f	Explain the role of correlation risk in market risk and credit risk.
P1	**g**	**Explain how correlation risk relates to systemic and concentration risk.**

WORKBOOK A8 + VARDEEZ FOCUS

▶ Vardeez: Concentration ratio: concept and calculation

A8·S5 concentration ratio concept and calculation, tied to the joint default probability (Q4)

MR-9 Empirical Properties of Correlation: How Do Correlations Behave in the Real World?

P3 ×2 P4 ×1 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *a) Identifying Best fit Distribution (Equity, Bond & default correlation) b) know calculation of mean reversion & autocorrelation concept*

RECALL WORKBOOK SECTIONS

D11 Empirical Properties of Correlation

Tier D • Map #54 (2/16) • 2 Qs • Vol 2 p.124 • LOs a to c

P4	a	Describe how equity correlations and correlation volatilities behave throughout various economic states.
P3	**b**	Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation.

WORKBOOK D11 + VARDEEZ FOCUS

▶ Vardeez: Mean reversion calculation and autocorrelation concept

D11·S2 mean reversion rate from regressing the correlation change on the lagged level

D11·S3 the corresponding autocorrelation (Q1)

P3	**c**	Identify the best-fit distribution for equity, bond, and default correlations.

WORKBOOK D11 + VARDEEZ FOCUS

▶ Vardeez: Best-fit distribution for equity, bond and default correlation

D11·S1 best-fit distribution for equity, bond and default probability correlations

MR-10 Financial Correlation Modeling – Bottom-Up Approaches

P2 ×1 P4 ×2 VARDEEZ: NOT SO IMP WORKBOOK BEST: TIER B

Vardeez: *Not so likely to be Tested*

! Not likely to be tested.

RECALL WORKBOOK SECTIONS

B3 Default Correlation, Joint PD and Copulas

Tier B • Map #16 (6/16) • TG ✓ • 3 Qs • Vol 2 p.24 • LOs a to c

P4 a Explain the purpose of copula functions and how they are applied in finance.

P2 **b** **Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets.**

WORKBOOK B3

B3·S4 Gaussian copula and implied correlation

P4 c Summarize the process of finding the default time of an asset correlated to all other assets in a portfolio using the Gaussian copula.

MR-11 Regression Hedging and Principal Component Analysis

P3 ×5 P4 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *a) Regression Hedge calculation (using the same formula, Beta as well as face value can be calculated)*

RECALL WORKBOOK SECTIONS

C1 Regression Hedging, DV01 and PCA

Tier C • Map #23 (5/16) • TG ✓ • 2 Qs • Vol 2 p.37 • LOs a to h

P3 **a** Explain the drawbacks to using a DV01-neutral hedge for a bond position.

WORKBOOK C1

C1·S3 DV01 hedging: the DV01-neutral amount as the planted distractor against the regression amount

C1·S4 drawbacks of a DV01-neutral hedge and why the regression hedge improves it

P3 **b** Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge.

WORKBOOK C1

C1·S4 drawbacks of a DV01-neutral hedge and why the regression hedge improves it

P3 **c** Calculate the regression hedge adjustment factor, beta.

WORKBOOK C1 + VARDEEZ FOCUS

▶ Vardeez: Regression hedge: beta

C1·S1 hedge ratio from a regression output that lists alpha, beta and both DV01s, with alpha as the tempting extra input

P3 **d** Calculate the face value of an offsetting position needed to carry out a regression hedge.

WORKBOOK C1 + VARDEEZ FOCUS

▶ Vardeez: Regression hedge: face value (same formula)

C1·S2 face amount of the offsetting position in a regression hedge, paired with the hedged P&L volatility

C1·S3 DV01 hedging: the DV01-neutral amount as the planted distractor against the regression amount

P4 e Calculate the face value of multiple offsetting swap positions needed to carry out a two-variable regression hedge.

P4 f Compare and contrast level and change regressions.

P4 g Explain why and how a regression hedge differs from a hedge based on a reverse regression.

P3 **h** Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.

WORKBOOK C1

C1·S5 PCA as the construction of a hedging portfolio: how many PCs two hedging securities can neutralize, and the risk-weight sum (Q2)

MR-12 Arbitrage Pricing with Term Structure Models

P3 ×3 P4 ×7 VARDEEZ: MODERATE WORKBOOK BEST: TIER C

Vardeez: *a) CMT swap*

RECALL WORKBOOK SECTIONS

C7 CMT Swap Valuation and OAS on a Rate Tree

Tier C • Map #33 (3/16) • 1 Qs • Vol 2 p.66 • LOs a to j

P4	a	Calculate the expected discounted value of a zero-coupon security using a binomial tree.
P4	b	Construct and apply an arbitrage argument to price a call option on a zero-coupon security using replicating portfolios.
P4	c	Define risk-neutral pricing and apply it to option pricing.
P3	**d**	Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. WORKBOOK C7 **C7·S3** true versus risk-neutral probabilities and the drift they imply, with the real-world probabilities supplied as a temptation (Q1)
P4	e	Explain how the principles of arbitrage pricing of derivatives on fixed-income securities can be extended over multiple periods.
P3	**f**	Define option-adjusted spread (OAS) and apply it to security pricing. WORKBOOK C7 **C7·S1** value a CMT swap backward through a two-period rate tree with risk-neutral probabilities, inside an OAS wrapper **C7·S2** OAS: shift the discount rates only, then read cheap versus rich from the sign
P4	g	Describe the rationale behind the use of recombining trees in option pricing.
P3	**h**	Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. WORKBOOK C7 + VARDEEZ FOCUS ► Vardeez: CMT swap **C7·S1** value a CMT swap backward through a two-period rate tree with risk-neutral probabilities, inside an OAS wrapper
P4	i	Evaluate the advantages and disadvantages of reducing the size of the time steps on the pricing of derivatives on fixed- income securities.
P4	j	Evaluate the appropriateness of the Black-Scholes-Merton model when valuing derivatives on fixed-income securities.

MR-13 Expectations, Risk Premium, Convexity, and the Shape of the Term Structure

P3 ×4 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *a) Jensen's inequality b) Remember this value convexity increases with bond volatility & time to maturity*

RECALL WORKBOOK SECTIONS

D12 Convexity and Jensen's Inequality

Tier D • Map #55 (2/16) • 1 Qs • Vol 2 p.128 • LOs a to d

D18 Inflation and Market Stress Across Bond Maturities

Tier D • Map #72 (2/16) • TG ✓ • 1 Qs • Vol 2 p.146 • LOs a to d

P3 **a** Explain the role of interest rate expectations in determining the shape of the term structure.

WORKBOOK D18

D18·S1 higher expected inflation as higher expected short rates: effect on forward rates and bond values grows with duration

D18·S3 a climate-related shock framed as an inflation and stress scenario for short versus long bonds (Q1)

P3 **b** Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

WORKBOOK D12, D18

D12·S2 the value of convexity increases with volatility

D18·S2 market stress as a higher risk premium and higher rate volatility: premium term grows with duration, convexity term pulls the other way

P3 **c** Estimate the convexity effect using Jensen's inequality.

WORKBOOK D12 + VARDEEZ FOCUS

► Vardeez: Jensen's inequality. Remember: the value of convexity increases with bond volatility and time to maturity

D12·S1 Jensen's inequality applied to bond prices: the expected price exceeds the price at the expected rate

D12·S2 the value of convexity increases with volatility

D12·S3 the value of convexity increases with time to maturity, so forward rates bend down more at longer terms (Q1)

P3 **d** Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor.

WORKBOOK D18

D18·S2 market stress as a higher risk premium and higher rate volatility: premium term grows with duration, convexity term pulls the other way

D18·S3 a climate-related shock framed as an inflation and stress scenario for short versus long bonds (Q1)

MR-14 The Art of Term Structure Models: Drift

P1 ×6 P3 ×1 P4 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *a) Vasicek Model Calc can come b) Ho-Lee Model c) Calc of Expected Rate in 7 years (LO 14.g) Note: Calculations are super imp here*

! Calculations are super important here.

RECALL WORKBOOK SECTIONS

A7 **Ho-Lee, Time-Dependent Drift and Volatility**

Tier A • Aug ✓ • Map #7 (7/16) • TG ✓ • 5 Qs • Vol 1 p.58 • LOs a to e

A25 **Gauss+ and Vasicek**

Tier A • Aug ✓ • Map #40 (2/16) • 5 Qs • Vol 1 p.178 • LOs h

D1 **CIR and Lognormal Short-Rate Models**

Tier D • Map #39 (4/16) • TG ✓ • 2 Qs • Vol 2 p.80 • LOs c, d, f

D12 **Convexity and Jensen's Inequality**

Tier D • Map #55 (2/16) • 1 Qs • Vol 2 p.128 • cross-reference

P1	a	**Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift.** WORKBOOK A7 **A7·S6** sibling swap against Vasicek and Models 1 and 2
P1	b	**Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift.** WORKBOOK A7 **A7·S6** sibling swap against Vasicek and Models 1 and 2
P3	c	Describe methods for addressing the possibility of negative short-term rates in term structure models. WORKBOOK D1 **D1·S3** non-negativity: CIR and lognormal versus normal models
P1	d	**Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift.** WORKBOOK A7 + VARDEEZ FOCUS ▶ Vardeez: Ho-Lee model **A7·S1** interest rate after 2 months in the top node of a monthly Ho-Lee tree **A7·S2** Ho-Lee and time-dependent volatility models: rate in the upper nodes **A7·S3** Ho-Lee and Model 3 two-step plug and play **A7·S4** no-arbitrage model, node-2 Δr and the drift read back from the nodes **A7·S5** two questions on the Ho-Lee short-rate tree **A7·S7** August plain forms: Ho-Lee plug values (Q1); time-dependent drift and volatility theory (Q5)
P1	e	**Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices.** WORKBOOK A7 **A7·S4** no-arbitrage model, node-2 Δr and the drift read back from the nodes
P1	f	**Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion.** WORKBOOK A7 + VARDEEZ FOCUS ▶ Vardeez: Vasicek model **A7·S6** sibling swap against Vasicek and Models 1 and 2
P1	g	**Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life.** WORKBOOK A25 + VARDEEZ FOCUS ▶ Vardeez: Vasicek calculations, incl. expected rate in T (e.g. 7) years (his "LO 14.g") **A25·S1** closed-book E[rT] = r0·e^(−kT) + θ(1 − e^(−kT)), with θ built from r∞ and the risk premium **A25·S2** closed-book half-life ln 2/k **A25·S3** Vasicek model calculation: rate change and standard deviation **A25·S4** LO 14.g expected rate in T years
P4	h	Describe the effectiveness of the Vasicek Model.

MR-15 The Art of Term Structure Models: Volatility and Distribution

P1 ×2 P2 ×3 P4 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *a) CIR model b) Concept of Terminal Distributions c) Model 3 d) Lognormal Model (Deterministic Drift) Understand Interpretation*

RECALL WORKBOOK SECTIONS

A7 Ho-Lee, Time-Dependent Drift and Volatility

Tier A • Aug ✓ • Map #7 (7/16) • TG ✓ • 5 Qs • Vol 1 p.58 • LOs a to c

D1 CIR and Lognormal Short-Rate Models

Tier D • Map #39 (4/16) • TG ✓ • 2 Qs • Vol 2 p.80 • LOs f

D18 Inflation and Market Stress Across Bond Maturities

Tier D • Map #72 (2/16) • TG ✓ • 1 Qs • Vol 2 p.146 • LOs d

P1 **a** **Describe the short-term rate process under a model with time-dependent volatility.**

WORKBOOK A7 + VARDEEZ FOCUS

▶ Vardeez: "Model 3" (time-dependent volatility)

A7·S2 Ho-Lee and time-dependent volatility models: rate in the upper nodes

A7·S7 August plain forms: Ho-Lee plug values (Q1); time-dependent drift and volatility theory (Q5)

P1 **b** **Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility.**

WORKBOOK A7 + VARDEEZ FOCUS

▶ Vardeez: "Model 3" calculation

A7·S2 Ho-Lee and time-dependent volatility models: rate in the upper nodes

A7·S3 Ho-Lee and Model 3 two-step plug and play

P4 c Assess the efficacy of time-dependent volatility models.

P2 **d** **Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models.**

WORKBOOK D1 + VARDEEZ FOCUS

▶ Vardeez: CIR model; concept of terminal distributions

D1·S3 non-negativity: CIR and lognormal versus normal models

D1·S5 the sibling set against Vasicek and Ho-Lee (Q2)

P2 **e** **Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models.**

WORKBOOK D1 + VARDEEZ FOCUS

▶ Vardeez: CIR and lognormal rate change

D1·S1 CIR model solving: one step of the short-rate change with drift, volatility and a random draw

D1·S2 CIR basis-point volatility σ_r versus the constant parameter σ

D1·S4 lognormal models: basis-point volatility proportional to r, with deterministic drift or with mean reversion

P2 **f** **Describe lognormal models with deterministic drift and mean reversion.**

WORKBOOK D1 + VARDEEZ FOCUS

▶ Vardeez: Lognormal model with deterministic drift: understand interpretation

D1·S4 lognormal models: basis-point volatility proportional to r, with deterministic drift or with mean reversion

MR-16 The Vasicek and Gauss+ Models

P1 ×3 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *New Chapter Highly Testable*

! New chapter, highly testable.

RECALL WORKBOOK SECTIONS

A25 Gauss+ and Vasicek

Tier A • Aug ✓ • Map #40 (2/16) • 5 Qs • Vol 1 p.178 • LOs a to d

P1 **a** **Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics.**

WORKBOOK A25 + VARDEEZ FOCUS

A25·S5 Gauss+ multi-factor structure: no shock on r, the $r \to m \to l \to \mu$ cascade, two sources of risk

P1 **b** **Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model.**

WORKBOOK A25 + VARDEEZ FOCUS

A25·S5 Gauss+ multi-factor structure: no shock on r, the $r \to m \to l \to \mu$ cascade, two sources of risk

P1 **c** **Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model.**

WORKBOOK A25 + VARDEEZ FOCUS

A25·S6 August plain form: Gauss+ factor changes (Q1 short rate, Q2 medium-term factor, Q3 long-term factor)

P2 **d** **Explain how the parameters of the Gauss+ model can be estimated from empirical data.**

VARDEEZ FOCUS

MR-17 Volatility Smiles

P1 ×5 P2 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Full chapter is IMP*

RECALL WORKBOOK SECTIONS

A3 Volatility Smiles, Skews and Put-Call Parity

Tier A • Aug ✓ • Map #4 (9/16) • TG ✓ • 4 Qs • Vol 1 p.30 • LOs a to i

P1 **a** **Describe a volatility smile and volatility skew.**

WORKBOOK A3 + VARDEEZ FOCUS

A3·S1 equity skew: effect of changing strike and maturity on implied volatility, plain form

P1 **b** **Explain the implications of put-call parity on the implied volatility of call and put options.**

WORKBOOK A3 + VARDEEZ FOCUS

A3·S7 put-call parity: a European call and put with the same strike and maturity share one implied volatility, tested in its smile context (Q4; distractor in Q3)

P1 **c** **Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset.**

WORKBOOK A3 + VARDEEZ FOCUS

A3·S4 lognormal vols versus implied vols for FX and index options

P1 **d** **Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility.**

WORKBOOK A3 + VARDEEZ FOCUS

A3·S2 two questions on FX and equity smiles

A3·S3 implied volatility shape for FX options

A3·S4 lognormal vols versus implied vols for FX and index options

P1 **e** **Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape.**

WORKBOOK A3 + VARDEEZ FOCUS

A3·S1 equity skew: effect of changing strike and maturity on implied volatility, plain form

A3·S2 two questions on FX and equity smiles

A3·S3 implied volatility shape for FX options

A3·S5 equity options volatility smile property

A3·S6 sibling swap: FX explanation attributed to equities or vice versa

P2 **f** **Describe alternative ways of characterizing the volatility smile.**

VARDEEZ FOCUS

P2 **g** **Describe volatility term structures and volatility surfaces and how they may be used to price options.**

VARDEEZ FOCUS

P2 **h** **Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures.**

VARDEEZ FOCUS

P2 **i** **Explain the impact of a single asset price jump on a volatility smile.**

VARDEEZ FOCUS

MR-18 Fundamental Review of the Trading Book

P1 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Again Full chapter is IMP (Refer GARP for Curve Risk)*

! Refer to the GARP book for curvature risk.

RECALL WORKBOOK SECTIONS

A4 Fundamental Review of the Trading Book (FRTB)

Tier A • Aug ✓ • Map #5 (8/16) • TG ✓ • 4 Qs • Vol 1 p.36 • LOs a to c

B2 Expected Shortfall and Coherence

Tier B • Map #12 (6/16) • TG ✓ • 3 Qs • Vol 2 p.19 • LOs a, b

C2 Backtesting VaR: Exceedance vs PIT

Tier C • Map #21 (4/16) • TG ✓ • 3 Qs • Vol 2 p.41 • LOs c

P1 **a** **Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes.**

WORKBOOK A4, B2 + VARDEEZ FOCUS

A4·S1 stressed ES at 97.5% replaces 10-day VaR at 99%, and why
A4·S5 FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes
A4·S6 curvature risk charge, from the GARP text (Q4)
B2·S2 ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces under FRTB

P1 **b** **Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons.**

WORKBOOK A4 + VARDEEZ FOCUS

A4·S2 liquidity horizons by asset class and the cascade ES

P1 **c** **Explain the FRTB revisions to Basel regulations in the following areas: - Classification of positions in the trading book compared to the banking book. - Backtesting, profit and loss attribution, credit risk, and securitizations.**

WORKBOOK A4 + VARDEEZ FOCUS

A4·S3 standardized vs internal models approach, with the desk-level backtesting and P&L attribution tests that decide which applies
A4·S4 trading book / banking book boundary

Credit Risk Measurement and Management

20% • 16 Qs

CR-1 Fundamentals of Credit Risk

P4 ×5 VARDEEZ: NOT SO IMP NO WORKBOOK SECTION

Vardeez: *Not Much Expected from this Chapter*

P4	a	Define credit risk and explain how it arises using examples.
P4	b	Explain the differences between insolvency, default, and bankruptcy.
P4	c	Identify and describe transactions that generate credit risk.
P4	d	Describe the entities that are exposed to credit risk and explain circumstances under which exposure occurs.
P4	e	Discuss the motivations for managing or taking on credit risk.

CR-2 Governance

P1 ×1 P3 ×1 P4 ×3 VARDEEZ: NOT SO IMP WORKBOOK BEST: TIER A

Vardeez: *Again Not much expected here. Just Read this Carefully (key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight)*

RECALL WORKBOOK SECTIONS

A22 Three Lines of Defence and Operational Risk Governance

Tier A • Aug ✓ • Map #29 (4/16) • TG ✓ • 3 Qs • Vol 1 p.161 • LOs a

P1	a	**Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control.** WORKBOOK A22 **A22·S1** three lines of operational risk management: who does what, with every distractor moving an activity across a line
P4	b	Explain the processes that lead to risk taking including credit origination, credit risk assessment, and credit approval processes.
P3	c	Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. VARDEEZ FOCUS ► Vardeez: Guidelines, Skills, Limits, Oversight: read carefully
P4	d	Describe the most common parameters of a credit-sensitive transaction.
P4	e	Describe the roles of the credit committee in an organization.

CR-3 Credit Risk Management

P3 ×3 P4 ×7 VARDEEZ: NOT SO IMP NO WORKBOOK SECTION

Vardeez: *Again not so imp, but read these 3 LOs carefully. LO 19.e & LO 19.g; and LO 19.h:*

! Vardeez cites these as ”LO 19.e, 19.g, 19.h”. The letters match CR-3 e, g, h exactly; ”19” is a different reading number (GARP global = 21).

P4	a	Describe key elements of an effective lending or financing policy.
P4	b	Explain the importance and challenges of setting exposure and concentration limits.
P4	c	Describe the scope and allocation processes of a bank's credit facility and explain bank-specific policies and actions to reduce credit risk.
P4	d	Discuss factors that should be considered during the credit asset classification process.
P3	**e**	Describe and explain loan loss provisions and loan loss reserves. VARDEEZ FOCUS ► Vardeez: Loan loss provisions and reserves
P4	f	Identify and explain the components of expected loss and differentiate between expected loss and unexpected loss.
P3	**g**	Explain the requirements for estimating expected loss under IFRS 9. VARDEEZ FOCUS ► Vardeez: IFRS 9 expected loss
P3	**h**	Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. VARDEEZ FOCUS ► Vardeez: Workout: retain vs write off loss assets
P4	i	Explain the components of credit risk analysis.
P4	j	Explain the components of credit risk management capacity, and identify key questions that the board of directors of a bank should ask.

CR-4 Capital Structure in Banks

P1 ×4 P2 ×1 P4 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Important Reading: unexpected Loss Calculations as well as risk contributions can show up*

RECALL WORKBOOK SECTIONS

A1 **Unexpected Loss, UL Contribution and Economic Capital**

Tier A • Aug ✓ • Map #1 (10/16) • TG ✓ • 4 Qs • Vol 1 p.13 • LOs a to i

D3 **Regulatory vs Economic Capital and Capital Adequacy**

Tier D • Map #61 (3/16) • TG ✓ • 2 Qs • Vol 2 p.89 • LOs i

P4 a Evaluate a bank's economic capital relative to its level of credit risk.

P4 b Identify and describe important factors used to calculate economic capital for credit risk: probability of default, exposure, and loss rate.

P2 **c Define and calculate expected loss (EL).**

VARDEEZ FOCUS

P1 **d Define and calculate unexpected loss (UL).**

WORKBOOK A1 + VARDEEZ FOCUS

▶ Vardeez: Unexpected loss calculation

A1·S1 UL for a single component: standalone UL from EA, PD, LR, σLR and the binomial σPD

A1·S5 intermediate-result trap: the single-asset UL planted when ULC is asked (Q2; standalone UL in Q1 and Q3)

P1 **e Estimate the variance of default probability assuming a binomial distribution.**

WORKBOOK A1 + VARDEEZ FOCUS

A1·S1 UL for a single component: standalone UL from EA, PD, LR, σLR and the binomial σPD

P1 **f Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size.**

WORKBOOK A1 + VARDEEZ FOCUS

▶ Vardeez: UL contributions (risk contributions)

A1·S2 ULC with all loans of the same value, ULC = UL × $\sqrt{\rho}$, plain form

A1·S3 portfolio UL for n identical loans, ULp = UL × $\sqrt{(n + \rho(n^2 - n))}$, forward and reverse-solved

A1·S5 intermediate-result trap: the single-asset UL planted when ULC is asked (Q2; standalone UL in Q1 and Q3)

P1 **g Describe how economic capital is derived.**

WORKBOOK A1, D3 + VARDEEZ FOCUS

A1·S4 economic capital EC = ULp × CM: the omitted capital multiplier

D3·S2 unexpected loss into economic capital: EC = ULp × CM, loan level ULCi × CM

P4 h Explain how the credit loss distribution is modeled.

P4 i Describe challenges to quantifying credit risk.

CR-5 Introduction to Credit Risk Modeling and Assessment

P1 ×3 P2 ×1 P3 ×1 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *LO 23.d is Imp, LO 23.e & LO 23.f: Imp (LO 23.g for RAROC is imp, but Raroc is more imp in Ops risk)*

RECALL WORKBOOK SECTIONS

A2 RAROC, Adjusted RAROC and Hurdle Rate

Tier A • Aug ✓ • Map #3 (9/16) • TG ✓ • 9 Qs • Vol 1 p.19 • LOs g

A5 Merton Model, Distance to Default and Risk-Neutral PD

Tier A • Aug ✓ • Map #2 (8/16) • TG ✓ • 6 Qs • Vol 1 p.43 • LOs e, f

A10 Altman Z-Score and Linear Discriminant Analysis

Tier A • Aug ✓ • Map #15 (6/16) • TG ✓ • 3 Qs • Vol 1 p.82 • LOs d

D3 Regulatory vs Economic Capital and Capital Adequacy

Tier D • Map #61 (3/16) • TG ✓ • 2 Qs • Vol 2 p.89 • LOs a to c

D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV

Tier D • Map #41 (2/16) • notes only • Vol 2 p.152 • LOs f

P4 a Explain the capital adequacy, asset quality, management, earnings, and liquidity (CAMEL) system used for evaluating the financial condition of a bank.

P4 b Describe quantitative measurements and factors of credit risk, including probability of default, loss given default, exposure at default, expected loss, and time horizon.

P3 **c** Estimate risk-weighted assets and capital adequacy ratio of a financial institution.

WORKBOOK D3

D3·S3 risk-weighted assets (standardized weights plus IRB K × 12.5 × EAD) and the capital adequacy ratio against a minimum (Q1)

P1 **d** **Describe the judgmental approaches, empirical models, and financial models to predict default.**

WORKBOOK A10 + VARDEEZ FOCUS

A10·S4 'calculate z score (LDA)': Z as Altman's linear discriminant function assigning a firm to a group

P1 **e** **Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model.**

WORKBOOK A5 + VARDEEZ FOCUS

▶ Vardeez: Merton model

A5·S1 distance to default, then assign a rating band from the score

A5·S6 short-form DtD ln(V /F)/σV versus the full form

A5·S7 scope trap: Merton over horizons beyond one year (Q6; two- and three-year horizons in Q1, Q2)

P2 **f** **Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model.**

WORKBOOK D20 + VARDEEZ FOCUS

D20·S1 CreditRisk+ as the actuarial, default-mode model (notes; distractors 1, 5)

D20·S4 Moody's KMV as the equity-market based Merton implementation (notes; distractor 4)

D20·S5 four-way comparison on loss recognition and correlation modelling (comparison table)

P1 **g** **Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan.**

WORKBOOK A2 + VARDEEZ FOCUS

▶ Vardeez: RAROC (more important in ORR-18)

A2·S1 compute after-tax RAROC, in plain form and with red-herring inputs

CR-6 Credit Scoring and Rating

P3 ×2 P4 ×2 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *CREDIT RATING AGENCY METHODOLOGIES (Just give a Good Read for This)*

RECALL WORKBOOK SECTIONS

D7 **Early Warning Indicators and PIT vs TTC**

Tier D • Map #45 (2/16) • TG ✓ • 2 Qs • Vol 2 p.106 • LOs a, b, d

D13 **Retail Credit Risk**

Tier D • Map #57 (2/16) • TG ✓ • 1 Qs • Vol 2 p.131 • LOs a, b

P4 a Compare the credit scoring system to the credit rating system in assessing credit quality and describe the different types of each system.

P3 **b** Differentiate between through-the-cycle credit rating system and point-in-time credit rating system.

WORKBOOK D7

D7·S3 PIT versus TTC definitions and users

D7·S4 the procyclicality polarity: PIT is the volatile, procyclical one (Q2)

P4 c Describe the process for developing credit risk scoring and rating models.

P3 **d** Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings.

VARDEEZ FOCUS

▶ Vardeez: Credit rating agency methodologies: give it a good read

CR-7 Credit Scoring and Retail Credit Risk Management

P3 ×4 P4 ×4 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *LO 25.f: & LO 25.h: can be important here*

RECALL WORKBOOK SECTIONS

D13 **Retail Credit Risk**

Tier D • Map #57 (2/16) • TG ✓ • 1 Qs • Vol 2 p.131 • LOs a to h

P4	a	Analyze the credit risks and other risks generated by retail banking.
P3	**b**	Explain the differences between retail credit risk and corporate credit risk. WORKBOOK D13 D13·S3 retail versus corporate credit risk: predictable EL, small but non-zero capital (Q1)
P3	**c**	Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. WORKBOOK D13 D13·S2 the dark side of retail credit: a systematic factor that drives many credits at once
P4	d	Define and describe credit risk scoring model types, key variables, and applications.
P4	e	Discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model.
P3	**f**	Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. VARDEEZ FOCUS ► Vardeez: CAP and accuracy ratio
P4	g	Describe the customer relationship cycle and discuss the trade-off between creditworthiness and profitability.
P3	**h**	Discuss the benefits of risk-based pricing of financial services. WORKBOOK D13 + VARDEEZ FOCUS ► Vardeez: Risk-based pricing D13·S1 risk-based pricing and adverse selection under a "one price fits all" policy; tiered pricing direction

CR-8 Country Risk: Determinants, Measures, and Implications

P3 ×1 P4 ×6 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *overall give a good read to this chapter. Same as Level 1, but Last LO is very imp on CDS*

! Give the whole chapter a good read; largely the same as Part I.

RECALL WORKBOOK SECTIONS

D23 **Sovereign Default Risk and CDS**

Tier D • Map #56 (1/16) • 1 Qs • Vol 2 p.162 • LOs a to g

P4	a	Identify and explain the different sources of country risk.
P4	b	Evaluate the methods for measuring country risk and discuss the limitations of using those methods.
P4	c	Compare and contrast foreign currency defaults and local currency defaults.
P4	d	Explain the consequences of a country's default.
P4	e	Discuss measures of sovereign default risk and describe components of a sovereign rating.
P4	f	Describe the shortcomings of the sovereign rating systems of rating agencies.
P3	**g**	Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. WORKBOOK D23 + VARDEEZ FOCUS ► Vardeez: Last LO is very important: CDS spreads D23·S1 sovereign CDS spreads as a measure of country default risk, compared with ratings and bond default spreads (CR-8 g) (Q1)

CR-9 Estimating Default Probabilities

P1 ×7 P2 ×5 P3 ×1 P4 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *MOST IMP CHAPTER. Almost all LOs are imp here, but do not focus on CDS here you can focus on this in later reading. Also expect questions on Hazard Rates, those marginal/conditional probabilities calculations, & also on merton and so on. (U can skip More Precise Hazard Rates)*

! MOST IMPORTANT CHAPTER in CR. Almost all LOs important.

RECALL WORKBOOK SECTIONS

A5 Merton Model, Distance to Default and Risk-Neutral PD

Tier A • Aug ✓ • Map #2 (8/16) • TG ✓ • 6 Qs • Vol 1 p.43 • LOs n

A10 Altman Z-Score and Linear Discriminant Analysis

Tier A • Aug ✓ • Map #15 (6/16) • TG ✓ • 3 Qs • Vol 1 p.82 • LOs a to c

A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD

Tier A • Aug ✓ • Map #17 (5/16) • TG ✓ • 7 Qs • Vol 1 p.113 • LOs k

D5 Credit Spread and PD from Spreads

Tier D • Map #42 (2/16) • 2 Qs • Vol 2 p.97 • LOs h, j, l

D14 Credit Derivatives and Credit-Linked Notes

Tier D • Map #58 (2/16) • 1 Qs • Vol 2 p.134 • LOs g

P2 **a** **Compare agencies' ratings to internal credit rating systems.**
VARDEEZ FOCUS

P1 **b** **Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality.**
WORKBOOK A10 + VARDEEZ FOCUS
A10·S1 direct plug: compute Z from a set of filings
A10·S2 the five ratios and their weights recalled closed-book, none given in the stem
A10·S3 cut-off zones used to classify a firm
A10·S4 'calculate z score (LDA)': Z as Altman's linear discriminant function assigning a firm to a group
A10·S5 the two LDAs: linear discriminant analysis for scoring versus the loss distribution approach for operational capital (Q2; the loss distribution approach itself is in D19)

P2 **c** **Describe the relationship between borrower rating and probability of default.**
VARDEEZ FOCUS

P1 **d** **Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default.**
WORKBOOK A15 + VARDEEZ FOCUS
▶ Vardeez: Marginal and conditional probability calculations
A15·S4 marginal PD as the difference of successive cumulative PDs

P1 **e** **Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities.**
WORKBOOK A5, A15 + VARDEEZ FOCUS
▶ Vardeez: Hazard rates (skip "more precise hazard rates")
A5·S4 Merton PD converted into an average hazard rate
A15·S1 hazard rate and PD: from a constant hazard rate to cumulative and year-t default probabilities
A15·S2 hazard rate from PD by the direct formula $\bar{\lambda} = -\ln(1-Q)/T$
A15·S5 sibling swap: $1 - e^{(-\lambda t)}$ is the cumulative PD, not the hazard rate
A15·S6 conditional versus unconditional default probability (Q1, Q2, Q3)

P2 **f** **Describe recovery rates and their dependencies on default rates.**
VARDEEZ FOCUS

P3 **g** Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller.
WORKBOOK D14
× Vardeez says skip: CDS: covered later (CR-13)
D14·S3 CDS mechanics: obligations of buyer and seller, settlement

P1 **h** **Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates.**
WORKBOOK A15
× Vardeez says skip: CDS: covered later
A15·S3 hazard rate from a CDS spread, $\bar{\lambda} = s/(1-R)$

P4 i Define and explain CDS-bond basis.
× Vardeez says skip: CDS: covered later

P2 **j** **Compare default probabilities calculated from historical data with those calculated from credit yield spreads.**
WORKBOOK D5 + VARDEEZ FOCUS
D5·S3 one-period PD = $(1/LGD) \times [1 - (1+rf)/(1+y)]$, where recovery decides the ranking (Q2)

P1 **k** **Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk.**
WORKBOOK A5 + VARDEEZ FOCUS
A5·S3 risk-neutral PD N(−d2) versus the real-world PD with μ

P1 **l** **Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model.**
WORKBOOK A5, D5 + VARDEEZ FOCUS
▶ Vardeez: Merton
A5·S2 back out V0 and σV from the simultaneous equations using the solver's converged values
A5·S5 leverage ratio to return on equity through the delta N(d1)
D5·S1 credit spread from the market value of debt, with the card's inverted sign as the distractor

P1 **m** **Calculate distance to default and default probability using the Merton model.**
WORKBOOK A5 + VARDEEZ FOCUS
▶ Vardeez: Merton distance to default
A5·S1 distance to default, then assign a rating band from the score
A5·S3 risk-neutral PD N(−d2) versus the real-world PD with μ
A5·S4 Merton PD converted into an average hazard rate
A5·S6 short-form DtD $\ln(V/F)/\sigma V$ versus the full form
A5·S7 scope trap: Merton over horizons beyond one year (Q6: two- and three-year horizons in Q1, Q2)

CR-10 Credit Value at Risk

P1 ×1 P2 ×3 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Here, We can expect Different Credit Risk Models comparison to show up on exam. Don't get into Formulas too much here, but atleast know the details of each of 4 Models discussed here. Credit Spread Risk can be imp too.*

RECALL WORKBOOK SECTIONS

A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD
Tier A • Aug ✓ • Map #17 (5/16) • TG ✓ • 7 Qs • Vol 1 p.113 • cross-reference

A31 Basel IRB Capital
Tier A • Aug ✓ • 5 Qs • Vol 1 p.214 • LOs d

B1 Credit VaR and Granularity
Tier B • Map #10 (6/16) • TG ✓ • 2 Qs • Vol 2 p.15 • LOs c, f

D5 Credit Spread and PD from Spreads
Tier D • Map #42 (2/16) • 2 Qs • Vol 2 p.97 • LOs f

D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV
Tier D • Map #41 (2/16) • notes only • Vol 2 p.152 • LOs d, e

P4 a Compare market risk value at risk (VaR) with credit VaR in terms of definition, time horizon, and tools for measuring them.

P2 **b Define and calculate Credit VaR.**
WORKBOOK B1
B1·S1 Credit VaR by direct formula: worst-case loss at the confidence level minus expected loss
B1·S2 formula-card form and the dropped-EL trap

P4 c Describe the use of rating transition matrices for calculating credit VaR.

P1 **d Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings- based (IRB) approach.**
WORKBOOK A31, D20 + VARDEEZ FOCUS
▶ Vardeez: Vasicek model: know the details
A31·S1 IRB capital for a single corporate loan under Advanced IRB, with the correlation formula and maturity adjustment: the plain form
D20·S3 Vasicek as the model under the Basel IRB formula (notes; distractors 3, 6)

P2 **e Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation.**
WORKBOOK D20 + VARDEEZ FOCUS
▶ Vardeez: Comparison of credit risk models: know each model, not formulas
D20·S1 CreditRisk+ as the actuarial, default-mode model (notes; distractors 1, 5)
D20·S2 CreditMetrics as the ratings-migration, mark-to-market model (notes; distractor 2)
D20·S3 Vasicek as the model under the Basel IRB formula (notes; distractors 3, 6)
D20·S5 four-way comparison on loss recognition and correlation modelling (comparison table)

P2 **f Define credit spread risk and assess its impact on calculating credit VaR.**
VARDEEZ FOCUS
▶ Vardeez: Credit spread risk

CR-11 Portfolio Credit Risk

P2 ×5 P4 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER B

Vardeez: *LO 29.a: LO 29.c: LO 29.h: (These 3 LOs are Very Important here) Any other LOs are not important. Single Factor model will be tested but not from this chapter - from later readings. Also, Just read Crash Course Notes for unimportant LOs. Conditional Default Probability can be imp, hence will upload a Practical Question for this on the Portal, just below this Particular Chapter.(Note : Do not get stuck on any Formulas or Notations)*

! Conditional default probability can be important (Vardeez posts a practice question).
! Do not get stuck on formulas or notation.

RECALL WORKBOOK SECTIONS

B1 Credit VaR and Granularity
Tier B • Map #10 (6/16) • TG ✓ • 2 Qs • Vol 2 p.15 • LOs c, g, h

B3 Default Correlation, Joint PD and Copulas

Tier B • Map #16 (6/16) • TG ✓ • 3 Qs • Vol 2 p.24 • LOs a to f

P2	a	**Define and calculate default correlation for credit portfolios.** WORKBOOK B3 + VARDEEZ FOCUS B3·S1 default correlation of two firms B3·S3 default correlation between credits with unequal default probabilities
P4	b	Identify drawbacks in using the correlation-based credit portfolio framework. × Vardeez says skip: Crash-course notes only
P2	c	**Assess the impact of correlation on a credit portfolio and its Credit VaR.** WORKBOOK B1 + VARDEEZ FOCUS B1·S4 granularity effect 2: it does not fall to zero when defaults are correlated
P2	d	**Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation.** WORKBOOK B3 × Vardeez says skip: Single factor model tested from later readings (CR-12 f), not here B3·S6 conditional default probability given the common factor
P4	e	Define beta and calculate the asset return correlation of any pair of firms using the single factor model. × Vardeez says skip: Crash-course notes only
P2	f	**Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model.** WORKBOOK B3 × Vardeez says skip: Crash-course notes only B3·S2 joint probability of default, direct calculation B3·S5 dropped-term trap: omitting $\pi 1 \times \pi 2$
P4	g	Describe how Credit VaR can be calculated using a simulation of joint defaults. × Vardeez says skip: Crash-course notes only
P2	h	**Assess the effect of granularity on Credit VaR.** WORKBOOK B1 + VARDEEZ FOCUS B1·S3 granularity effect 1: Credit VaR falls as the portfolio becomes more granular B1·S4 granularity effect 2: it does not fall to zero when defaults are correlated

CR-12 Credit Risk

P2 ×1 P3 ×3 P4 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER B

Vardeez: *LO30.g: This LO which talks abt Credit Var is imp for exam. Also here LO 30.f, in this we have One Factor Model I was talking abt. Imp it is for the exam. LO 30.e here we can consider it relevant for the exam. So read it carefully. Rest all other LOs are just a repeat and not so relevant. just skim through*

! Rest are repeats; skim.

RECALL WORKBOOK SECTIONS

B1 Credit VaR and Granularity
Tier B • Map #10 (6/16) • TG ✓ • 2 Qs • Vol 2 p.15 • LOs g

B3 Default Correlation, Joint PD and Copulas
Tier B • Map #16 (6/16) • TG ✓ • 3 Qs • Vol 2 p.24 • LOs e, f

C8 Counterparty Exposure Profiles: EE, PFE, EPE
Tier C • Map #37 (3/16) • TG ✓ • 2 Qs • Vol 2 p.69 • LOs a

D5 Credit Spread and PD from Spreads
Tier D • Map #42 (2/16) • 2 Qs • Vol 2 p.97 • LOs c

D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV
Tier D • Map #41 (2/16) • notes only • Vol 2 p.152 • LOs g

P4	a	Assess the credit risks of derivatives.
P4	b	Define credit valuation adjustment (CVA) and debt valuation adjustment (DVA).
P3	**c**	Calculate the probability of default using credit spreads. WORKBOOK D5 D5·S1 credit spread from the market value of debt, with the card's inverted sign as the distractor D5·S2 deriving PD from credit spreads( ) D5·S3 one-period PD = (1/LGD) × [1 − (1 + rf)/(1 + y)], where recovery decides the ranking (Q2)
P4	d	Describe, compare, and contrast various credit risk mitigants and their role in credit analysis.
P3	**e**	Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. VARDEEZ FOCUS ► Vardeez: Read carefully
P2	**f**	**Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model.** WORKBOOK B3, D15 + VARDEEZ FOCUS ► Vardeez: One-factor model: important B3·S4 Gaussian copula and implied correlation D15·S4 Gaussian copula and implied correlation
P3	**g**	Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. WORKBOOK D20 + VARDEEZ FOCUS ► Vardeez: Credit VaR D20·S2 CreditMetrics as the ratings-migration, mark-to-market model (notes; distractor 2)

CR-13 Credit Derivatives

P3 ×4 P4 ×4 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *LO 31.b - Just know the Process here - This can be very lengthy on the exam especially calculations. So just KNOW the process. And (LO 31.g: This can be imp too) (LO 31.f: Know this as well)*

RECALL WORKBOOK SECTIONS

D14 **Credit Derivatives and Credit-Linked Notes**

Tier D • Map #58 (2/16) • 1 Qs • Vol 2 p.134 • LOs a to h

P3	**a**	Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). WORKBOOK D14 D14·S1 CLN structure: funded, the note investor bears the credit risk D14·S2 total return swaps: who owns the bond, who earns the spread D14·S3 CDS mechanics: obligations of buyer and seller, settlement
P3	**b**	Explain how to account for credit risk exposure in valuing a CDS. VARDEEZ FOCUS ► Vardeez: Know the PROCESS only; exam calculations can be lengthy
P4	c	Identify the default probabilities used to value a CDS.
P4	d	Evaluate the use of credit indices and fixed coupons in pricing CDS transactions.
P4	e	Define CDS forwards and CDS options.
P3	**f**	Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. WORKBOOK D14 + VARDEEZ FOCUS D14·S4 synthetic CDO valuation and the two implied correlations, compound and base (Q1, option C)
P3	**g**	Define the two measures of implied correlation: compound (tranche) correlation and base correlation. WORKBOOK D14 + VARDEEZ FOCUS D14·S4 synthetic CDO valuation and the two implied correlations, compound and base (Q1, option C)
P4	h	Discuss alternative approaches used to estimate default correlation.

CR-14 Derivatives

P3 ×1 P4 ×9 VARDEEZ: NOT SO IMP WORKBOOK BEST: TIER B

Vardeez: *LO 32.i - Counterparty Risk Intermediaries - These can be imp from exam point of view.*

RECALL WORKBOOK SECTIONS

B2 Expected Shortfall and Coherence

Tier B • Map #12 (6/16) • TG ✓ • 3 Qs • Vol 2 p.19 • LOs j

P4	a	Define derivatives and explain how derivative transactions create counterparty credit risk.
P4	b	Compare and contrast exchange-traded derivatives and over-the-counter (OTC) derivatives, and discuss the features of their markets.
P4	c	Describe the process of clearing a derivative transaction.
P4	d	Identify the participants and describe the use of collateralization in the derivatives market.
P4	e	Define the International Swaps and Derivatives Association (ISDA) Master Agreement, the risk-mitigating features it provides, and the default events it covers.
P4	f	Describe the features and use of credit derivatives and discuss potential risks they may create.
P4	g	Describe central clearing of OTC derivatives and discuss the roles, mandate, advantages, and disadvantages of the central counterparty (CCP).
P4	h	Explain the margin requirements for both centrally-cleared and non-centrally-cleared derivatives.
P3	i	Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. VARDEEZ FOCUS ► Vardeez: Counterparty risk intermediaries
P4	j	Describe the approaches used and the challenges faced in modeling derivatives risk.

CR-15 Counterparty Risk and Beyond

P3 ×8 VARDEEZ: MODERATE NO WORKBOOK SECTION

Vardeez: *This is the Foundational for Counterparty Risk Chapters. So give a good read and understand the points.*

! Foundation for the counterparty risk chapters. Give it a good read and understand the points.

P3	**a**	Describe counterparty risk and differentiate it from lending risk. VARDEEZ FOCUS
P3	**b**	Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. VARDEEZ FOCUS
P3	**c**	Identify and describe institutions that take on significant counterparty risk. VARDEEZ FOCUS
P3	**d**	Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. VARDEEZ FOCUS
P3	**e**	Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. VARDEEZ FOCUS
P3	**f**	Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. VARDEEZ FOCUS
P3	**g**	Identify and explain the costs of an OTC derivative. VARDEEZ FOCUS
P3	**h**	Explain the components of the X-Value Adjustment (xVA) term. VARDEEZ FOCUS

CR-16 Netting, Close-out and Related Aspects

P1 ×1 P2 ×2 P4 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Termination Features & trade compression- These seem to be crucial here.*

RECALL WORKBOOK SECTIONS

A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting

Tier A • Aug ✓ • Map #27 (4/16) • TG ✓ • 6 Qs • Vol 1 p.152 • LOs a to f

P4	a	Explain the purpose of an International Swaps and Derivatives Association (ISDA) master agreement.
P4	b	Summarize netting and close-out procedures (including multilateral netting), explain their advantages and disadvantages, and describe how they fit into the framework of the ISDA master agreement.
P4	c	Describe the effectiveness of netting in reducing credit exposure under various scenarios.
P1	d	**Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages.** WORKBOOK A21 + VARDEEZ FOCUS ▶ Vardeez: Termination features and trade compression **A21·S7** termination features and trade compression, CR-16 d (Reading 34)
P2	e	**Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression.** VARDEEZ FOCUS ▶ Vardeez: Trade compression
P2	f	**Identify and describe termination events and discuss their potential effects on parties to a transaction.** VARDEEZ FOCUS ▶ Vardeez: Termination events

CR-17 Margin (Collateral) and Settlement

P1 ×6 P4 ×5 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *(LO 35.g - Know the terms here very well). LO 35.j This too can be imp. (LO 35.f - This is really imp abt disputes in collateral)*

RECALL WORKBOOK SECTIONS

A20 Wrong-Way and Right-Way Risk

Tier A • Aug ✓ • Map #26 (4/16) • TG ✓ • 2 Qs • Vol 1 p.147 • LOs e, j

A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting

Tier A • Aug ✓ • Map #27 (4/16) • TG ✓ • 6 Qs • Vol 1 p.152 • LOs a to k

P4	a	Describe the rationale for collateral management.
P1	**b**	**Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount.** WORKBOOK A21 **A21·S3** overcollateralisation: initial margin and haircuts (Q5); the securitisation overcollateralisation account is drilled in A9
P1	**c**	**Calculate the credit support amount (margin) under various scenarios.** WORKBOOK A21 **A21·S1** collateral required from the MtM value, threshold, MTA, rounding and margin already held **A21·S2** the threshold-plus-MTA test that fails, so no transfer is made
P4	d	Describe the role of a valuation agent.
P4	e	Describe the mechanics of collateral and the types of collateral that are typically used.
P1	**f**	**Explain the process for the reconciliation of collateral disputes.** WORKBOOK A21 + VARDEEZ FOCUS ▶ Vardeez: Collateral disputes: really important **A21·S5** disputes and reconciliation, CR-17 f (LO 35.f)
P1	**g**	**Explain the features of a collateralization agreement.** WORKBOOK A21 + VARDEEZ FOCUS ▶ Vardeez: Know the terms very well **A21·S4** features and terms of a collateral agreement, CR-17 g (LO 35.g)
P1	**h**	**Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality.** WORKBOOK A21 **A21·S8** two-way vs one-way CSA and rating-linked terms, as seen in August
P4	i	Explain aspects of collateral including funding, rehypothecation, and segregation.
P1	**j**	**Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization.** WORKBOOK A21 + VARDEEZ FOCUS **A21·S6** market, operational and liquidity risk through collateralisation, with role misassignment, CR-17 j (LO 35.j)
P4	k	Describe the various regulatory capital requirements.

CR-18 Central Clearing

P3 ×1 P4 ×9 VARDEEZ: MODERATE WORKBOOK BEST: TIER D

Vardeez: *CCP is same as Level 1. Not likely to be testable. Just give a good read. (Know Loss Waterfall)*

! CCP is the same as Part I; not likely testable. Just give a good read.

RECALL WORKBOOK SECTIONS

D24 **Central Clearing and the CCP Loss Waterfall**

Tier D • Map #59 (1/16) • 1 Qs • Vol 2 p.165 • LOs a to j

P4	a	Define a central counterparty (CCP) and describe the mechanics of central clearing.
P4	b	Explain the concept of novation under central clearing.
P4	c	Define netting, multilateral offset, and compression and provide examples of each.
P4	d	Describe the application and estimation of margin and default funds under central clearing.
P4	e	Discuss the risks faced by a CCP and the ways it manages its exposures.
P3	**f**	Provide examples of a loss waterfall. WORKBOOK D24 + VARDEEZ FOCUS ► Vardeez: Know the loss waterfall D24·S1 the order of the loss waterfall, with sequence errors as distractors (Q1)
P4	g	Explain the different methods of absorbing losses and managing the default of one or more members of a CCP.
P4	h	Compare bilateral and central clearing.
P4	i	Compare initial margin and default fund requirements for clearing members in relation to loss coverage, cost of clearing, and moral hazard.
P4	j	Describe the advantages and disadvantages of central clearing.

CR-19 Future Value and Exposure

P1 ×1 P2 ×4 P3 ×1 P4 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *Various Payoff profiles are important. (This one SECURITY EXPOSURE PROFILES). Pls do this. After this Netting Factor Formula and impact is important. Pls go to Crash Course for this. (LO 36.g: Just read this as well)*

! Vardeez cites ”LO 36.g”. The text matches CR-19 g (funding vs credit exposure); GARP global number = 37.

RECALL WORKBOOK SECTIONS

C8 **Counterparty Exposure Profiles: EE, PFE, EPE**

Tier C • Map #37 (3/16) • TG ✓ • 2 Qs • Vol 2 p.69 • LOs a to i

P3	**a**	Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. WORKBOOK C8 **C8·S1** the six-way metric family: expected MtM (EFV), expected exposure / EPE, PFE, ENE, effective EE/EPE and average EPE, maximum PFE
P4	b	Compare the characterization of credit exposure to VaR methods and describe additional considerations used in the determination of credit exposure.
P4	c	Identify factors that affect the calculation of the credit exposure profile and summarize the impact of collateral on exposure.
P2	**d**	**Identify typical credit exposure profiles for various derivative contracts and combination profiles.** WORKBOOK C8 + VARDEEZ FOCUS ▶ Vardeez: Payoff / security exposure profiles **C8·S3** sold option red herring: zero exposure whatever Black-Scholes data the stem supplies (Q1)
P2	**e**	**Explain how payment frequencies and exercise dates affect the exposure profile of various securities.** VARDEEZ FOCUS ▶ Vardeez: Security exposure profiles
P2	**f**	**Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values.** WORKBOOK C8 + VARDEEZ FOCUS ▶ Vardeez: Netting factor formula and impact (see crash course) **C8·S2** the netting row: aggregation, EFV additive but EPE not, netting factor
P2	**g**	**Describe the differences between funding exposure and credit exposure.** VARDEEZ FOCUS ▶ Vardeez: Just read
P1	**h**	**Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount.** WORKBOOK A21 **A21·S9** margin period of risk: components and effect on collateralised exposure and CVA, as seen in August (Q3, Q5)
P4	i	Assess the impact of collateral on counterparty risk and funding, with and without segregation or rehypothecation.

CR-20 CVA

P1 ×12 P2 ×5 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *CVA/BCVA and DVA is a favorite topic of GARP. Know the formulas very well. You cannot afford to skip this chapter at all.*

! CVA / BCVA / DVA is a GARP favourite. Know the formulas very well. Cannot skip.

RECALL WORKBOOK SECTIONS

A6 CVA, DVA, BCVA and CVA as a Spread
Tier A • Aug ✓ • Map #6 (8/16) • TG ✓ • 5 Qs • Vol 1 p.51 • LOs a to i

A20 Wrong-Way and Right-Way Risk
Tier A • Aug ✓ • Map #26 (4/16) • TG ✓ • 2 Qs • Vol 1 p.147 • LOs p

A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting
Tier A • Aug ✓ • Map #27 (4/16) • TG ✓ • 6 Qs • Vol 1 p.152 • LOs j

P2 **a** **Explain the motivation for and the challenges of pricing counterparty risk.**
VARDEEZ FOCUS

P2 **b** **Describe credit value adjustment (CVA).**
VARDEEZ FOCUS

P1 **c** **Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization.**
WORKBOOK A6 + VARDEEZ FOCUS
A6·S1 UCVA from an EPE profile and a PD term structure, LGD × EPE × PD
A6·S2 CVA as a running spread: −average EPE × spread, and upfront CVA divided by the risky annuity

P2 **d** **Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA.**
VARDEEZ FOCUS

P1 **e** **Describe debt value adjustment (DVA) and bilateral CVA (BCVA).**
WORKBOOK A6 + VARDEEZ FOCUS
A6·S3 BCVA = CVA + DVA with the sign convention CVA negative, DVA positive

P1 **f** **Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA.**
WORKBOOK A6 + VARDEEZ FOCUS
A6·S6 change from UCVA to BCVA, with the BCVA itself planted as a distractor (Q5)

P1 **g** **Calculate DVA, BCVA, and BCVA as a spread.**
WORKBOOK A6 + VARDEEZ FOCUS
A6·S3 BCVA = CVA + DVA with the sign convention CVA negative, DVA positive
A6·S4 BCVA with survival (first-to-default) adjustments
A6·S5 BCVA as a spread, plain form as seen in August

P2 **h** **Explain how netting can be incorporated into the CVA calculation.**
VARDEEZ FOCUS

P1 **i** **Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread.**
WORKBOOK A6 + VARDEEZ FOCUS
A6·S2 CVA as a running spread: −average EPE × spread, and upfront CVA divided by the risky annuity

P1 **j** **Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins.**
WORKBOOK A21 + VARDEEZ FOCUS
A21·S9 margin period of risk: components and effect on collateralised exposure and CVA, as seen in August (Q3, Q5)

P1 **k** **Describe wrong-way risk and contrast it with right-way risk.**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S1 classify trades as wrong-way or right-way, plain form as seen in August
A20·S2 the test: is the exposure linked to the counterparty's credit quality, and in which direction

P1 **l** **Identify examples of wrong-way risk and examples of right-way risk.**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S1 classify trades as wrong-way or right-way, plain form as seen in August

P1 **m** **Discuss the impact of collateral on wrong-way risk.**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S3 wrong-way collateral

P1 **n** **Identify examples of wrong-way collateral.**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S3 wrong-way collateral

P1 **o** **Discuss the impact of wrong-way risk on central counterparties (CCPs).**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S4 the effect of WWR on CCPs

P1 **p** **Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches.**
WORKBOOK A20 + VARDEEZ FOCUS
A20·S5 modelling approaches: hazard rate (intensity), structural, parametric, jump (Q2)

P2 **q** **Explain the implications of central clearing on wrong-way risk.**
VARDEEZ FOCUS

CR-21 The Evolution of Stress Testing Counterparty Exposures

P2 ×8 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *Full Important Chapter. Stress Testing has always been favorite topic of Garp.*

! Fully important chapter. Stress testing is a GARP favourite.

RECALL WORKBOOK SECTIONS

C9 Stress-Testing Practices and Scenario Design

Tier C • Map #38 (3/16) • TG ✓ • 2 Qs • Vol 2 p.74 • LOs a to h

P2 **a** **Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure.**

VARDEEZ FOCUS

P2 **b** **Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution.**

VARDEEZ FOCUS

P2 **c** **Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio.**

VARDEEZ FOCUS

P2 **d** **Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio.**

VARDEEZ FOCUS

P2 **e** **Describe a stress test that can be performed on CVA.**

VARDEEZ FOCUS

P2 **f** **Calculate the stressed CVA and the stress loss on CVA.**

VARDEEZ FOCUS

P2 **g** **Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR.**

VARDEEZ FOCUS

P2 **h** **Describe the common pitfalls in stress testing CCR.**

WORKBOOK C9 + VARDEEZ FOCUS

C9·S1 appropriate stress-testing practices as a best-practice audit: what to stress, how to aggregate, what to avoid

CR-22 Structured Credit Risk

P1 ×3 P2 ×2 P3 ×1 P4 ×7 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *LO 40.i - This is Imp LO. Just remember the impacts via the table. There can be one calculation question come from this reading. However Would recommend do it at the very end coz that will just consume your time. Concept of LO 40.L - Implied correlation seems likely to be testable.*

! Do this reading at the very end; it consumes time.

RECALL WORKBOOK SECTIONS

A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios

Tier A • Aug ✓ • Map #9 (7/16) • TG ✓ • 7 Qs • Vol 1 p.72 • LOs a to m

B3 Default Correlation, Joint PD and Copulas

Tier B • Map #16 (6/16) • TG ✓ • 3 Qs • Vol 2 p.24 • LOs l

D15 Structured Credit and Implied Correlation

Tier D • Map #60 (2/16) • TG ✓ • 1 Qs • Vol 2 p.137 • LOs l

P4	a	Describe common types of structured products.
P4	b	Describe tranching and the distribution of credit losses in a securitization.
P1	**c**	**Describe a waterfall structure in a securitization.** WORKBOOK A9 **A9·S4** shares paid to the senior and equity tranches in the final-year waterfall
P4	d	Identify the key participants in the securitization process and describe conflicts of interest that can arise in the process.
P1	**e**	**Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.** WORKBOOK A9 **A9·S2** cash flow into the equity tranche after the interim waterfall, with defaults and fees **A9·S4** shares paid to the senior and equity tranches in the final-year waterfall
P1	**f**	**Describe the treatment of excess spread in a securitization structure and estimate the value of the over-collateralization account at the end of each year.** WORKBOOK A9 **A9·S3** overcollateralisation account balance at a year end
P4	g	Explain the tests on the excess spread that a custodian must go through at the end of each year to determine the cash flow to the overcollateralization account and to the equity noteholders.
P3	**h**	Describe a simulation approach to calculating credit losses for different tranches in a securitization. WORKBOOK D15 **D15·S2** the one possible calculation: reading tranche values and writedowns off the simulation grid
P2	**i**	**Explain how the default probabilities and default correlations affect the credit risk in a securitization.** WORKBOOK B3, D15 + VARDEEZ FOCUS ► Vardeez: Remember the impacts via the table; one calculation question possible **B3·S7** LO 40.i impacts table for tranches (D15.1) **D15·S1** LO 40.i impacts table: how default probability and correlation move each tranche
P4	j	Explain how default sensitivities for tranches are measured.
P4	k	Describe risk factors that impact structured products.
P2	**l**	**Define implied correlation and describe how it can be measured.** WORKBOOK B3, D15 + VARDEEZ FOCUS ► Vardeez: Implied correlation: likely testable **B3·S4** Gaussian copula and implied correlation **D15·S3** implied correlation backed out of a tranche price **D15·S4** Gaussian copula and implied correlation
P4	m	Identify the motivations for using structured credit products.

CR-23 An Introduction to Securitisation

P1 ×2 P3 ×1 P4 ×5 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Don't expect much from this chapter on the exam. Not difficult chapter, almost same as Level 1. But LO 39.c: This can be important. (LO 39.g: Some calculations pls check them out.) Debt Service Coverage Ratio, SMM, CPR*

! Vardeez cites "LO 39.c / 39.g". The text matches CR-23 c and g; GARP global number = 41.

RECALL WORKBOOK SECTIONS

A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios

Tier A • Aug ✓ • Map #9 (7/16) • TG ✓ • 7 Qs • Vol 1 p.72 • LOs a to h

D14 Credit Derivatives and Credit-Linked Notes

Tier D • Map #58 (2/16) • 1 Qs • Vol 2 p.134 • cross-reference

P4	a	Define securitization, describe the securitization process, and explain the role of participants in the process.
P4	b	Explain the terms over-collateralization, first-loss piece, equity piece, and cash waterfall within the securitization process.
P3	**c**	Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. VARDEEZ FOCUS ▶ Vardeez: Trust vs SPV
P4	d	Explain the reasons for and the benefits of undertaking securitization.
P4	e	Describe and assess the various types of credit enhancements.
P4	f	Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to.
P1	**g**	**Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures.** WORKBOOK A9 + VARDEEZ FOCUS ▶ Vardeez: Calculations: DSCR and the ratios **A9·S1** delinquency ratio, plain numerical as seen in August **A9·S7** closed-book formulas CPR, DSCR and WAL (Q6, Q7)
P1	**h**	**Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate.** WORKBOOK A9 + VARDEEZ FOCUS ▶ Vardeez: SMM, CPR **A9·S5** CPR, SMM and PSA conversion, with the deal's overcollateralisation as the twist **A9·S6** prepayment rate theory and forecasting methodology **A9·S7** closed-book formulas CPR, DSCR and WAL (Q6, Q7)

Operational Risk and Resilience

20% • 16 Qs

Vardeez's ORR theme list (one cell in the sheet, spanning several readings). Mapped: 1 → ORR-10/11/12/13 • 2 → ORR-14 • 3 → ORR-21/22/23 • 4 → ORR-17 • 5 → ORR-24 b (no LO names LDA; workbook D19 sits on ORR-4 f) • 6 → EVT and block maxima sit in MR-3 • 7, 9 → ORR-22 a • 8 → ORR-18.

1. TPRM, AML & Outsourcing; 2. Investor Protection (MiFID & Volcker Rule); 3. Basel III - Three Components & Comparison with Basel I & II; 4. Pre-SCAP vs. Post-SCAP; 5. LDA Approach; 6. EVT & Block Maxima (Comparison); 7. Stressed Market VaR (Calculations); 8. RAROC (Calculations). 9) Basel 2.5 Market Risk Calcualtion.

ORR-1 to ORR-9: Vardeez says crash course notes only; no tier given.

ORR-1 Introduction to Operational Risk and Resilience

P1 ×2 P4 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A22 Three Lines of Defence and Operational Risk Governance

Tier A • Aug ✓ • Map #29 (4/16) • TG ✓ • 3 Qs • Vol 1 p.161 • LOs b

A23 Cyber Risk and Operational Resilience

Tier A • Aug ✓ • Map #31 (4/16) • TG ✓ • 2 Qs • Vol 1 p.167 • LOs a, c, d

P4 a Describe an operational risk management framework and assess the types of risks that can fall within the scope of such a framework.

P1 **b Describe the seven Basel II event risk categories and identify examples of operational risk events in each category.**

WORKBOOK A22

A22·S2 Basel event-type taxonomy, Level 1, Level 2 and the Level 3 examples

P4 c Explain characteristics of operational risk exposures and operational loss events, and challenges that can arise in managing operational risk due to these characteristics.

P1 **d Describe operational resilience, identify the elements of an operational resilience framework, and summarize regulatory expectations for operational resilience.**

WORKBOOK A23

A23·S2 operational resiliency as a regulatory concept: UK first, then BCBS, ECB, MAS

A23·S3 applied vulnerability comparison: payment bank versus community bank during COVID (Q2)

ORR-2 Risk Governance

P1 ×3 P4 ×1 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A22 Three Lines of Defence and Operational Risk Governance

Tier A • Aug ✓ • Map #29 (4/16) • TG ✓ • 3 Qs • Vol 1 p.161 • LOs a to d

P4 a Explain the Basel regulatory expectations for the governance of an operational risk management framework.

P1 **b Describe and compare the roles of different committees and the board of directors in operational risk governance.**

WORKBOOK A22

A22·S3 committees versus the board, and who owns risk appetite and limits (Q2)

P1 **c Describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense.**

WORKBOOK A22

A22·S1 three lines of operational risk management: who does what, with every distractor moving an activity across a line

P1 **d Explain best practices and regulatory expectations for the development of a risk appetite for operational risk and for a strong risk culture.**

WORKBOOK A22

A22·S3 committees versus the board, and who owns risk appetite and limits (Q2)

ORR-3 Risk Identification

P1 ×1 P4 ×3 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A22 Three Lines of Defence and Operational Risk Governance
Tier A • Aug ✓ • Map #29 (4/16) • TG ✓ • 3 Qs • Vol 1 p.161 • LOs c, d

C6 Model Risk Management
Tier C • Map #32 (3/16) • TG ✓ • 2 Qs • Vol 2 p.61 • LOs c

D2 Outsourcing and Third-Party Risk
Tier D • Map #51 (3/16) • TG ✓ • 2 Qs • Vol 2 p.84 • LOs d

D21 Investor Protection, MiFID and the Volcker Rule
Tier D • Map #52 (2/16) • notes only • Vol 2 p.155 • LOs d

D22 Operational Risk Identification, Mitigation and Reporting
Tier D • Map #75 (2/16) • notes only • Vol 2 p.158 • LOs a to d

P4 a Discuss different top-down and bottom-up approaches and tools for identifying operational risks.

P4 b Describe best practices in extreme risk identification for operational risk.

P4 c Describe and apply an operational risk taxonomy and give examples of different taxonomies of operational risks.

P1 **d Describe and apply the Level 1, 2, and 3 categories in the Basel operational risk taxonomy.**

WORKBOOK A22

A22·S2 Basel event-type taxonomy, Level 1, Level 2 and the Level 3 examples

ORR-4 Risk Measurement and Assessment

P1 ×2 P3 ×3 P4 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A17 Extreme Value Theory: GEV vs POT
Tier A • Aug ✓ • Map #25 (5/16) • TG ✓ • 2 Qs • Vol 1 p.127 • LOs f

D2 Outsourcing and Third-Party Risk
Tier D • Map #51 (3/16) • TG ✓ • 2 Qs • Vol 2 p.84 • LOs g

D19 LDA for Operational Risk
Tier D • Map #74 (2/16) • 1 Qs • Vol 2 p.149 • LOs a, f

D22 Operational Risk Identification, Mitigation and Reporting
Tier D • Map #75 (2/16) • notes only • Vol 2 p.158 • LOs a to g

P3	**a**	Explain best practices for the collection of operational loss data and reporting of operational loss incidents, including regulatory expectations. WORKBOOK D22 D22·S4 loss-data collection: regulatory criteria, fields, dates, boundary events, completeness
P3	**b**	Explain operational risk-assessment processes and tools, including risk control self-assessments (RCSAs), likelihood assessment scales, and heatmaps. WORKBOOK D22 D22·S2 risk and control self-assessment (RCSA): inherent and residual risk, scales, heatmap, ownership of notes outcomes
P3	**c**	Describe the differences among key risk indicators (KRIs), key performance indicators (KPIs), and key control indicators (KCIs). WORKBOOK D22 D22·S3 key risk indicators, against KPIs and KCIs
P4	d	Describe the use of factor-based models that quantitatively assess operational risk, and explain the application of the Swiss cheese model and the bowtie tool.
P4	e	Estimate operational risk exposures based on the fault tree model given probability assumptions.
P1	**f**	**Describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations.** WORKBOOK A10, A17, D19 A10·S5 the two LDAs: linear discriminant analysis for scoring versus the loss distribution approach for operational capital (Q2; the loss distribution approach itself is in D19) A17·S6 operational-risk framing of EVT and block maxima D19·S1 loss frequency × severity convolution into an aggregated annual loss distribution D19·S2 operational risk capital as the 99.9th percentile of the aggregated distribution, with aggregation across units of measure
P1	**g**	**Describe and explain the steps to ensure a strong level of operational resilience, and to test the operational resilience of important business services.** WORKBOOK A23 A23·S2 operational resiliency as a regulatory concept: UK first, then BCBS, ECB, MAS

ORR-5 Risk Mitigation

P3 ×2 P4 ×5 VARDEEZ: UNTIERED WORKBOOK BEST: TIER D

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

D2 Outsourcing and Third-Party Risk
Tier D • Map #51 (3/16) • TG ✓ • 2 Qs • Vol 2 p.84 • LOs g

D21 Investor Protection, MiFID and the Volcker Rule
Tier D • Map #52 (2/16) • notes only • Vol 2 p.155 • LOs d

D22 Operational Risk Identification, Mitigation and Reporting
Tier D • Map #75 (2/16) • notes only • Vol 2 p.158 • LOs a to g

P4	a	Explain and compare different ways firms address their operational risk exposures.
P3	**b**	Compare different types of internal controls and provide examples of each type of internal control. WORKBOOK D22 D22·S1 the control sibling set: preventive, detective, corrective, directive, with key controls, automation and notes testing
P3	**c**	Describe control automation, internal control design, and control testing, including risks and challenges that arise in these processes and ways to make them more effective. WORKBOOK D22 D22·S1 the control sibling set: preventive, detective, corrective, directive, with key controls, automation and notes testing
P4	d	Describe methods to improve the quality of an operational process and reduce the potential for human error.
P4	e	Explain how operational risk can arise with new products, new business initiatives, or mergers and acquisitions, and describe ways to mitigate these risks.
P4	f	Identify and describe approaches firms should use to mitigate the impact of operational risk events.
P4	g	Describe methods for the transfer of operational risks and the management of reputational risk, and assess their effectiveness in different situations.

ORR-6 Risk Reporting

P4 ×4 VARDEEZ: UNTIERED WORKBOOK BEST: TIER D

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

D19 LDA for Operational Risk

Tier D • Map #74 (2/16) • 1 Qs • Vol 2 p.149 • LOs c

D22 Operational Risk Identification, Mitigation and Reporting

Tier D • Map #75 (2/16) • notes only • Vol 2 p.158 • LOs a to d

P4	a	Identify roles and responsibilities of different organizational committees, and explain how risk reports should be developed for each committee or business function.
P4	b	Describe components of operational risk reports and explain best practices in operational risk reporting.
P4	c	Describe challenges to reporting operational risks, including characteristics of operational loss data, and explain ways to overcome these challenges.
P4	d	Explain best practices for reporting risk exposures to regulators and external stakeholders.

ORR-7 Integrated Risk Management

P3 ×1 P4 ×3 VARDEEZ: UNTIERED NO WORKBOOK SECTION

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

P4	a	Describe the role of risk governance, risk appetite, and risk culture in the context of an enterprise risk management (ERM) framework.
P3	**b**	Explain and differentiate between regulatory capital and economic capital requirements as prescribed in Basel regulations. WORKBOOK D3 D3·S1 regulatory capital as a formula-driven minimum versus economic capital as the bank's own estimate
P4	c	Describe the elements of a sound stress-testing framework for financial institutions and explain best practices for stress testing.
P4	d	Explain challenges and considerations when developing and implementing models used in stress testing operational risk.

ORR-8 Cyber-resilience: Range of Practices

P1 ×2 P3 ×1 P4 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A23 Cyber Risk and Operational Resilience

Tier A • Aug ✓ • Map #31 (4/16) • TG ✓ • 2 Qs • Vol 1 p.167 • LOs a to e

P4	a	Define cyber-resilience and compare recent regulatory initiatives in the area of cyber-resilience.
P1	**b**	**Describe current practices by banks and supervisors in the governance of a cyber-risk management framework, including roles and responsibilities.** WORKBOOK A23 **A23·S1** cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons
P1	**c**	**Explain methods for supervising cyber-resilience, testing and incident response approaches, and cybersecurity and resilience metrics.** WORKBOOK A23 **A23·S1** cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons
P4	d	Explain and assess current practices for the sharing of cybersecurity information between different types of institutions.
P3	**e**	Describe practices for the governance of risks of interconnected third-party service providers. WORKBOOK D2 D2·S2 concentration risk in a critical service provider and its resilience implications

ORR-9 Cyberthreats and Information Security Risks

P1 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Vardeez: use crash course notes for ORR-1 to ORR-9; no Schweser needed.

RECALL WORKBOOK SECTIONS

A23 Cyber Risk and Operational Resilience

Tier A • Aug ✓ • Map #31 (4/16) • TG ✓ • 2 Qs • Vol 1 p.167 • LOs a, b

P1 **a** **Provide examples of cyber threats and information security risks, and describe frameworks and best practices for managing cyber risks.**

WORKBOOK A23

A23·S1 cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons

P1 **b** **Describe lessons learned from the Equifax case study.**

WORKBOOK A23

A23·S1 cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons

ORR-10 Sound Management of Risks related to Money Laundering and Financing of Terrorism

P1 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

! Theme: TPRM, AML and outsourcing.

RECALL WORKBOOK SECTIONS

A29 Money Laundering and Financing of Terrorism

Tier A • Aug ✓ • Map #50 (2/16) • TG ✓ • 1 Qs • Vol 1 p.205 • LOs a to c

P1 **a** **Explain best practices recommended by the Basel committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks.**

WORKBOOK A29 + VARDEEZ FOCUS

A29·S1 ML/FT governance role trap: who files suspicious transaction reports and contacts supervisors and FIUs, and what the chief AML/CFT officer may and may not combine

A29·S4 the three stages of laundering: placement, layering, integration (Q1)

P1 **b** **Describe recommended practices for the acceptance, verification, and identification of customers at a bank.**

WORKBOOK A29 + VARDEEZ FOCUS

A29·S2 customer due diligence, including reliance on another bank's CDD

P1 **c** **Explain practices for managing ML/FT risks in a group-wide and cross-border context.**

WORKBOOK A29 + VARDEEZ FOCUS

A29·S3 correspondent banking risk

ORR-11 Financial Crime and Fraud

P1 ×1 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

! Theme: AML and fraud controls.

RECALL WORKBOOK SECTIONS

A29 Money Laundering and Financing of Terrorism

Tier A • Aug ✓ • Map #50 (2/16) • TG ✓ • 1 Qs • Vol 1 p.205 • LOs a, b

P1 **a** **Describe elements of a control framework to manage financial fraud risk and money laundering risk.**

WORKBOOK A29 + VARDEEZ FOCUS

A29·S4 the three stages of laundering: placement, layering, integration (Q1)

P2 **b** **Summarize the regulatory findings and describe the lessons learned from the USAA case study.**

VARDEEZ FOCUS

ORR-12 Guidance on Managing Outsourcing Risk

P2 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

! Theme: TPRM, AML and outsourcing.

RECALL WORKBOOK SECTIONS

D2 **Outsourcing and Third-Party Risk**

Tier D • Map #51 (3/16) • TG ✓ • 2 Qs • Vol 2 p.84 • LOs a to c

D21 **Investor Protection, MiFID and the Volcker Rule**

Tier D • Map #52 (2/16) • notes only • Vol 2 p.155 • LOs a

P2 **a** **Explain how risks can arise through outsourcing activities to third-party service providers and describe elements of an effective program to manage outsourcing risk.**

WORKBOOK D2 + VARDEEZ FOCUS

D2·S1 outsource the activity, never the accountability, applied to a vendor and subcontractor failure

P2 **b** **Explain how financial institutions should perform due diligence on third-party service providers.**

VARDEEZ FOCUS

P2 **c** **Describe topics and provisions that should be addressed in a contract with a third-party service provider.**

VARDEEZ FOCUS

ORR-13 Third-Party Risk Management

P2 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

! Theme: TPRM, AML and outsourcing.

RECALL WORKBOOK SECTIONS

D2 **Outsourcing and Third-Party Risk**

Tier D • Map #51 (3/16) • TG ✓ • 2 Qs • Vol 2 p.84 • LOs a, b

P2 **a** **Explain how risks related to the use of third parties can arise and describe characteristics of an effective third-party risk management framework.**

WORKBOOK D2 + VARDEEZ FOCUS

D2·S2 concentration risk in a critical service provider and its resilience implications

P2 **b** **Describe the lessons learned from the presented case studies.**

VARDEEZ FOCUS

ORR-14 Investor Protection and Compliance Risks in Investment Activities

P2 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

RECALL WORKBOOK SECTIONS

D21 **Investor Protection, MiFID and the Volcker Rule**

Tier D • Map #52 (2/16) • notes only • Vol 2 p.155 • LOs a, b

P2 **a** **Summarize important regulations designed to protect investors in financial instruments, including MiFiD, MiFiD II, and Dodd-Frank.**

WORKBOOK D21 + VARDEEZ FOCUS

▶ Vardeez: MiFID, Volcker Rule (investor protection)

D21·S1 suitability and appropriateness obligations

D21·S3 conflicts-of-interest management under MiFID II and MiFIR

D21·S4 the Volcker Rule's proprietary-trading prohibition and its market-making and hedging exemptions, notes with scope/condition errors

P2 **b** **Describe lessons learned from the case studies involving violations of investor protection or compliance regulations.**

WORKBOOK D21 + VARDEEZ FOCUS

D21·S2 best execution, with the Deutsche Bank Securities case

ORR-15 Supervisory Guidance on Model Risk Management

P2 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

RECALL WORKBOOK SECTIONS

C6 Model Risk Management

Tier C • Map #32 (3/16) • TG ✓ • 2 Qs • Vol 2 p.61 • LOs a to d

P2 **a** **Describe model risk and explain how it can arise in the implementation of a model.**

VARDEEZ FOCUS

P2 **b** **Describe elements of an effective model risk management process.**

WORKBOOK C6 + VARDEEZ FOCUS

C6·S2 the three components of model risk management: development, implementation and use; validation; governance, policies and controls

P2 **c** **Explain best practices for the development and implementation of models.**

VARDEEZ FOCUS

P2 **d** **Describe elements of a strong model validation process and challenges to an effective validation process.**

WORKBOOK C6 + VARDEEZ FOCUS

C6·S3 the independence requirement for validation, with effective challenge (incentives, competence, influence)

ORR-16 Case Study: Model Risk and Model Validation

P2 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

RECALL WORKBOOK SECTIONS

C6 Model Risk Management

Tier C • Map #32 (3/16) • TG ✓ • 2 Qs • Vol 2 p.61 • LOs a to c

P2 **a** **Define a model and describe different ways that financial institutions can become exposed to model risk.**

VARDEEZ FOCUS

P2 **b** **Describe the role of the model risk management function and explain best practices in the model risk management and validation processes.**

WORKBOOK C6 + VARDEEZ FOCUS

C6·S3 the independence requirement for validation, with effective challenge (incentives, competence, influence)

P2 **c** **Describe lessons learned from the three case studies involving model risk.**

WORKBOOK C6 + VARDEEZ FOCUS

C6·S1 the friction between investor and rating agency labelled model error, tied to the GARP Gaussian copula case

ORR-17 Stress Testing Banks

P3 ×2 P4 ×1 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER C

RECALL WORKBOOK SECTIONS

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes

Tier C • Map #30 (4/16) • TG ✓ • 2 Qs • Vol 2 p.56 • LOs a

C9 Stress-Testing Practices and Scenario Design

Tier C • Map #38 (3/16) • TG ✓ • 2 Qs • Vol 2 p.74 • LOs a to c

P3	**a**	Describe the evolution of the stress testing process and compare the methodologies of historical European Banking Association (EBA), Comprehensive Capital Analysis and Review (CCAR), and Supervisory Capital Assessment Program (SCAP) stress tests. WORKBOOK C5, C9 + VARDEEZ FOCUS ► Vardeez: Pre-SCAP vs post-SCAP **C5·S1** pre-SCAP versus post-SCAP features of stress testing **C5·S5** the SCAP / CEBS / EBA / CCAR sibling swap: which banks failed, base versus stress disclosure, which risk weights (Q2) **C9·S3** the SCAP / EBA / CCAR sibling swap on which regime did what (Q2; regime comparison also in C5.2)
P3	**b**	Explain challenges in designing stress test scenarios, including the problem of coherence in modeling risk factors. WORKBOOK C9 **C9·S1** appropriate stress-testing practices as a best-practice audit: what to stress, how to aggregate, what to avoid
P4	c	Explain challenges in modeling a bank's revenues, losses, and its balance sheet over a stress test horizon period.

ORR-18 Risk Capital Attribution and Risk-Adjusted Performance Measurement

P1 ×3 P3 ×1 P4 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

RECALL WORKBOOK SECTIONS

A2 RAROC, Adjusted RAROC and Hurdle Rate
Tier A • Aug ✓ • Map #3 (9/16) • TG ✓ • 9 Qs • Vol 1 p.19 • LOs a to h

D3 Regulatory vs Economic Capital and Capital Adequacy
Tier D • Map #61 (3/16) • TG ✓ • 2 Qs • Vol 2 p.89 • LOs a

D7 Early Warning Indicators and PIT vs TTC
Tier D • Map #45 (2/16) • TG ✓ • 2 Qs • Vol 2 p.106 • LOs d

P3	**a**	Define, compare, and contrast risk capital, economic capital, and regulatory capital and explain methods and motivations for using economic capital approaches to allocate risk capital. WORKBOOK D3 **D3·S1** regulatory capital as a formula-driven minimum versus economic capital as the bank's own estimate
P4	b	Describe the RAROC (risk-adjusted return on capital) methodology and its use in capital budgeting.
P1	**c**	**Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance.** WORKBOOK A2 + VARDEEZ FOCUS ► Vardeez: RAROC calculations **A2·S1** compute after-tax RAROC, in plain form and with red-herring inputs **A2·S3** direction of the effect on RAROC of changes in economic capital, income and expense
P4	d	Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level.
P1	**e**	**Calculate the hurdle rate and apply this rate in making business decisions using RAROC.** WORKBOOK A2 + VARDEEZ FOCUS ► Vardeez: Hurdle rate **A2·S4** find the hurdle rate: market-value-weighted cost of common equity (CAPM) and preferred equity **A2·S5** weighted cost of capital given break-even cost (Q8; break-even revenue at the hurdle in Q3)
P1	**f**	**Calculate the adjusted RAROC for a project to determine its viability.** WORKBOOK A2 + VARDEEZ FOCUS ► Vardeez: Adjusted RAROC **A2·S2** two on adjusted RAROC: value plus accept/reject decision, and the break-even beta
P4	g	Explain challenges in modeling diversification benefits, including aggregating a firm's risk capital and allocating economic capital to different business lines.
P4	h	Explain best practices in implementing an approach that uses RAROC to allocate economic capital.

ORR-19 Range of Practices and Issues in Economic Capital Frameworks

P4 ×4 VARDEEZ: NOT SO IMP WORKBOOK BEST: TIER B

RECALL WORKBOOK SECTIONS

B2 Expected Shortfall and Coherence

Tier B • Map #12 (6/16) • TG ✓ • 3 Qs • Vol 2 p.19 • LOs a

P4	a	Within the economic capital implementation framework, describe the challenges that appear in: - Defining and calculating risk measures - Risk aggregation - Validation of models - Dependency modeling in credit risk - Evaluating counterparty credit risk - Assessing interest rate risk in the banking book
P4	b	Describe the recommendations by the Bank for International Settlements (BIS) that supervisors should consider in order to make effective use of internal risk measures, such as economic capital, that are not designed for regulatory purposes.
P4	c	Explain benefits and impacts of using an economic capital framework within the following areas: - Credit portfolio management - Risk-based pricing - Customer profitability analysis - Management incentives
P4	d	Describe best practices and assess key concerns for the governance of an economic capital framework.

ORR-20 Capital Planning at Large Bank Holding Companies

P3 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

RECALL WORKBOOK SECTIONS

D10 BHC Capital Planning and VaR Model Validation

Tier D • Map #53 (2/16) • TG ✓ • 2 Qs • Vol 2 p.119 • LOs a, b

P3	a	Describe the Federal Reserve's Capital Plan Rule and explain the seven principles of an effective capital adequacy process for bank holding companies (BHCs) subject to the Capital Plan Rule. WORKBOOK D10 D10·S1 capital planning governance: board versus senior management, reporting of overlays
P3	b	Describe practices that can result in a strong and effective capital adequacy process for a BHC in the following areas: - Risk identification - Internal controls, including model review and validation - Corporate governance - Capital policy, including setting of goals and targets and contingency planning - Stress testing and stress scenario design - Estimating losses, revenues, and expenses, including quantitative and qualitative methodologies - Assessing the impact of capital adequacy, including risk-weighted asset (RWA) and balance sheet projections WORKBOOK D10 D10·S1 capital planning governance: board versus senior management, reporting of overlays D10·S2 BHC scenario design and estimation practices that the Federal Reserve calls stronger or weaker

ORR-21 Capital Regulation Before the Global Financial Crisis

P1 ×3 P3 ×3 P4 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Untiered in the sheet, but his theme "Basel III: three components and comparison with Basel I and II" draws on this reading.

RECALL WORKBOOK SECTIONS

A4 Fundamental Review of the Trading Book (FRTB)

Tier A • Aug ✓ • Map #5 (8/16) • TG ✓ • 4 Qs • Vol 1 p.36 • LOs c

A31 Basel IRB Capital

Tier A • Aug ✓ • 5 Qs • Vol 1 p.214 • LOs e, f

C2 Backtesting VaR: Exceedance vs PIT

Tier C • Map #21 (4/16) • TG ✓ • 3 Qs • Vol 2 p.41 • LOs c

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes

Tier C • Map #30 (4/16) • TG ✓ • 2 Qs • Vol 2 p.56 • LOs d, g, h

D4 Stressed VaR and Market Risk Capital

Tier D • Map #73 (3/16) • TG ✓ • 2 Qs • Vol 2 p.93 • LOs c, f

P3	a	Explain the motivations for introducing the Basel regulations, including key risk exposures addressed, and explain the reasons for revisions to Basel regulations over time. WORKBOOK C5 C5·S4 the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord
P3	b	Explain the calculation of risk-weighted assets and the capital requirement per the original Basel I guidelines. VARDEEZ FOCUS ► Vardeez: Basel I (for the Basel III comparison)
P1	c	**Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book.** WORKBOOK A4, C5, D4 A4·S5 FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes C5·S4 the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord D4·S2 market risk capital charge calculation: max(VaR(t−1), m × VaR(avg)), √10 scaling, multiplier from backtesting
P3	d	Describe changes to the Basel regulations made as part of Basel II, including the three pillars. WORKBOOK C5 + VARDEEZ FOCUS ► Vardeez: Basel II (for the Basel III comparison) C5·S4 the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord
P1	e	**Compare the standardized internal ratings-based (IRB) approach, the foundation IRB approach, and the advanced IRB approach for the calculation of credit risk capital under Basel II.** WORKBOOK A31 A31·S3 Foundation versus Advanced IRB inputs: supervisory LGD and the PD floor
P1	f	**Calculate credit risk capital under Basel II utilizing the IRB approach.** WORKBOOK A31, D4 A31·S1 IRB capital for a single corporate loan under Advanced IRB, with the correlation formula and maturity adjustment: the plain form A31·S2 retail IRB: fixed correlation, no maturity adjustment, reported as RWA D4·S3 mixed vignette: market stressed VaR charge plus credit capital given the 99.9th percentile default rate, one error per option (Q2)
P4	g	Compare the basic indicator approach, the standardized approach, and the advanced measurement approach for the calculation of operational risk capital under Basel II.
P4	h	Summarize elements of the Solvency II capital framework for insurance companies.

ORR-22 Solvency, Liquidity and Other Regulation After the Global Financial Crisis

P1 ×2 P3 ×5 P4 ×1 VARDEEZ: UNTIERED WORKBOOK BEST: TIER A

! Tier cell in the sheet reads ”0”. His themes ”Stressed market VaR (calculations)” and ”Basel 2.5 market risk calculation” land squarely here. Treat as high priority.

RECALL WORKBOOK SECTIONS

A4 Fundamental Review of the Trading Book (FRTB)
Tier A • Aug ✓ • Map #5 (8/16) • TG ✓ • 4 Qs • Vol 1 p.36 • LOs a to c

A31 Basel IRB Capital
Tier A • Aug ✓ • 5 Qs • Vol 1 p.214 • LOs h

A32 Net Stable Funding Ratio, LCR and Leverage Ratio
Tier A • Aug ✓ • 3 Qs • Vol 1 p.221 • LOs f

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes
Tier C • Map #30 (4/16) • TG ✓ • 2 Qs • Vol 2 p.56 • LOs a, d, e, h

D4 Stressed VaR and Market Risk Capital
Tier D • Map #73 (3/16) • TG ✓ • 2 Qs • Vol 2 p.93 • LOs a to c

D25 Contingent Convertible Bonds
Tier D • Map #70 (1/16) • TG ✓ • 1 Qs • Vol 2 p.168 • LOs g

P1 **a** **Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge.**

WORKBOOK A4, C5, D4 + VARDEEZ FOCUS

▶ Vardeez: Stressed VaR calculation; Basel 2.5 market risk calculation

A4·S5 FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes

C5·S4 the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord

D4·S1 stressed market VaR calculation: the SVaR term with its own max and multiplier

D4·S2 market risk capital charge calculation: max(VaR(t−1), m × VaR(avg)), √10 scaling, multiplier from backtesting

D4·S3 mixed vignette: market stressed VaR charge plus credit capital given the 99.9th percentile default rate, one error per option (Q2)

P3 **b** Explain the process of calculating the incremental risk capital charge for positions held in a bank's trading book.

VARDEEZ FOCUS

▶ Vardeez: Basel 2.5

P3 **c** Describe the comprehensive risk (CR) capital charge for portfolios of positions that are sensitive to correlations between default risks.

VARDEEZ FOCUS

▶ Vardeez: Basel 2.5

P3 **d** Define in the context of Basel III and calculate where appropriate: - Tier 1 capital and its components - Tier 2 capital and its components - Required Tier 1 equity capital, total Tier 1 capital, and total capital

VARDEEZ FOCUS

▶ Vardeez: Basel III components

P3 **e** Describe the motivations for and calculate the capital conservation buffer and the countercyclical buffer, including special rules for globally systemically important banks (G-SIBs).

VARDEEZ FOCUS

▶ Vardeez: Basel III buffers

P1 **f** **Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio.**

WORKBOOK A32 + VARDEEZ FOCUS

▶ Vardeez: Basel III liquidity and leverage ratios

A32·S1 NSFR from a balance sheet with ASF and RSF factors given: the plain form

A32·S2 NSFR shortfall: new funding needed to reach 100%, with both the funding leg and the deployment leg

A32·S3 NSFR versus LCR with both parameter sets in one table: pick the right factors (Q3)

P3 **g** Describe the mechanics of contingent convertible bonds (CoCos) and explain the motivations for banks to issue them.

WORKBOOK D25

D25·S1 trigger mechanics: Core Tier 1 to RWA threshold or regulator's nonviability declaration, and the 5.125% line between Additional Tier 1 and Tier 2

D25·S2 equity conversion versus principal write-down: both increase equity

D25·S3 AT1 ranking and the inversion controversy: CoCo holders bear losses not so different from equity holders (Q1)

P4 h Provide examples of legislative and regulatory reforms that were introduced after the 2007-2009 financial crisis.

ORR-23 High-level Summary of Basel III Reforms

P1 ×2 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

RECALL WORKBOOK SECTIONS

A31 Basel IRB Capital

Tier A • Aug ✓ • 5 Qs • Vol 1 p.214 • LOs a to c

A32 Net Stable Funding Ratio, LCR and Leverage Ratio

Tier A • Aug ✓ • 3 Qs • Vol 1 p.221 • LOs b

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes

Tier C • Map #30 (4/16) • TG ✓ • 2 Qs • Vol 2 p.56 • LOs a to c

P2 **a** **Explain the motivations for revising the Basel III framework and the goals and impacts of the December 2017 reforms to the Basel III framework.**

VARDEEZ FOCUS

P1 **b** **Summarize the December 2017 revisions to the Basel III framework in the following areas: - The standardized approach to credit risk - The internal ratings-based (IRB) approaches for credit risk - The CVA risk framework - The operational risk framework - The leverage ratio framework**

WORKBOOK A31, C5 + VARDEEZ FOCUS

A31·S4 December 2017 IRB revisions: A-IRB removed for certain asset classes, equities out of IRB, input floors

C5·S2 the standardised versus internal-models split across accords: who uses which, and where the December 2017 reforms removed models

P1 **c** **Describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor.**

WORKBOOK A31, C5 + VARDEEZ FOCUS

A31·S5 the 72.5% output floor and how input floors and the output floor bind (Q5)

C5·S3 output floors: the Basel II floor on Basel I and the Basel III floor on the revised standardised approaches

ORR-24 Basel III: Finalising Post-Crisis Reforms

P3 ×2 P4 ×1 VARDEEZ: UNTIERED WORKBOOK BEST: TIER C

! Untiered. His theme ”LDA approach” is not named in any 2026 LO; the closest anchor is ORR-24 b (AMA comparison) and ORR-4 f.

RECALL WORKBOOK SECTIONS

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes

Tier C • Map #30 (4/16) • TG ✓ • 2 Qs • Vol 2 p.56 • LOs a to c

P4 a Explain the elements of the new standardized approach to measure operational risk capital, including the business indicator, internal loss multiplier, and loss component, and calculate the operational risk capital requirement for a bank using this approach.

P3 **b** Compare the Standardized Measurement Approach (SMA) to earlier methods of calculating operational risk capital, including the Advanced Measurement Approaches (AMA).

VARDEEZ FOCUS

▶ Vardeez: SMA vs AMA (where LDA sits)

P3 **c** Describe general and specific criteria recommended by the Basel Committee for the identification, collection, and treatment of operational loss data.

WORKBOOK D22

D22·S4 loss-data collection: regulatory criteria, fields, dates, boundary events, completeness

Liquidity and Treasury Risk Measurement and Management 15% • 12 Qs

LTR-1 Liquidity Risk

P1 ×1 P3 ×2 P4 ×1 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Calculation of liquidity trading risk LVAR calculation. Second is liquidity black holes and positive feedback traders*

RECALL WORKBOOK SECTIONS

A11 Liquidity-Adjusted VaR and Cost of Liquidation

Tier A • Aug ✓ • Map #11 (6/16) • TG ✓ • 5 Qs • Vol 1 p.87 • LOs a

D27 Intraday Liquidity and Dealer Bank Failure Mechanics

Tier D • Map #65 (1/16) • notes only • Vol 2 p.174 • LOs b

P1 **a** **Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).**

WORKBOOK A11 + VARDEEZ FOCUS

► Vardeez: LVaR calculation (liquidity trading risk)

A11·S1 LVaR numerical with the normal-market (constant-spread) cost, the 'CAR, LVaR' pairing (CAR itself: see D3)

A11·S2 stressed (exogenous-spread) cost of liquidation 0.5 V (μ + zσ), with the constant-spread input as the sibling-swap distractor

A11·S3 cost of liquidation (CoL) as its own number, plain form of the August recall

A11·S4 LVaR theory: normal vs stressed spread, perfect spread correlation, liquidation horizon

P3 **b** Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft.

WORKBOOK D27

D27·S4 case↔lesson matching against Northern Rock (notes: case table, LTR-1 b)

P4 c Evaluate Basel III liquidity risk ratios and BIS principles for sound liquidity risk management.

P3 **d** Explain liquidity black holes and identify the causes of positive feedback trading.

VARDEEZ FOCUS

► Vardeez: Liquidity black holes and positive feedback traders

LTR-2 Liquidity and Leverage

P1 ×1 P4 ×6 VARDEEZ: NOT SO IMP WORKBOOK BEST: TIER A

Vardeez: *Calculation of expected transaction cost*

RECALL WORKBOOK SECTIONS

A11 Liquidity-Adjusted VaR and Cost of Liquidation

Tier A • Aug ✓ • Map #11 (6/16) • TG ✓ • 5 Qs • Vol 1 p.87 • LOs e, f

P4 a Differentiate between sources of liquidity risk and describe specific challenges faced by different types of financial institutions in managing liquidity risk.

P4 b Summarize the asset-liability management process at a fractional reserve bank, including the process of liquidity transformation.

P4 c Compare transactions used in the collateral market and explain risks that can arise through collateral market transactions.

P4 d Describe the relationship between leverage and a firm's return profile (including the leverage effect), and explain the impact of different types of transactions on a firm's leverage and balance sheet.

P4 e Describe and compare methods to measure and manage funding liquidity risk and transactions liquidity risk.

P1 **f** **Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.**

WORKBOOK A11 + VARDEEZ FOCUS

► Vardeez: Expected transaction cost calculation

A11·S5 Malz expected transactions cost and 99% spread risk factor

A11·S6 Malz multi-day liquidation adjustment, with time to escape from trading volume (Q4)

P4 g Discuss interactions between different types of liquidity risk and explain how liquidity risk events can increase systemic risk.

LTR-3 Early Warning Indicators

P2 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *Full chapter is important*

RECALL WORKBOOK SECTIONS

D7 Early Warning Indicators and PIT vs TTC

Tier D • Map #45 (2/16) • TG ✓ • 2 Qs • Vol 2 p.106 • LOs a to c

P2 **a** **Evaluate the characteristics of sound Early Warning Indicators (EWI) measures.**

WORKBOOK D7 + VARDEEZ FOCUS

D7·S1 characteristics of sound EWIs: leading, sharp, calibrated thresholds

P2 **b** **Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve).**

VARDEEZ FOCUS

P2 **c** **Discuss the applications of EWIs in the context of the liquidity risk management process.**

WORKBOOK D7 + VARDEEZ FOCUS

D7·S2 applications of EWIs in the liquidity risk process: escalation to the right authority

LTR-4 The Investment Function in Financial-Services Management

P1 ×2 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Money market investments learning objective 69C apply investment maturity strategies*

RECALL WORKBOOK SECTIONS

A30 Asset vs Borrowed Liquidity Management and the Investment Function

Tier A • Aug ✓ • Map #63 (1/16) • 2 Qs • Vol 1 p.209 • LOs a to c

P1 **a** **Compare various money market and capital market instruments and discuss their advantages and disadvantages.**

WORKBOOK A30 + VARDEEZ FOCUS

▶ Vardeez: Money market investments

A30·S3 investment instruments available to banks and their advantages and disadvantages

P1 **b** **Identify and discuss various factors that affect the choice of investment securities by a bank.**

WORKBOOK A30

A30·S4 factors affecting the choice of securities, including tax exposure (Q2)

P2 **c** **Apply investment maturity strategies and maturity management tools based on the yield curve and duration.**

VARDEEZ FOCUS

▶ Vardeez: Investment maturity strategies (his ”LO 69.c”)

LTR-5 Liquidity and Reserves Management: Strategies and Policies

P1 ×3 VARDEEZ: MODERATE WORKBOOK BEST: TIER A

Vardeez: *L O seventy A and L O seventy B*

RECALL WORKBOOK SECTIONS

A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing

Tier A • Aug ✓ • Map #64 (2/16) • 4 Qs • Vol 1 p.185 • LOs c

A30 Asset vs Borrowed Liquidity Management and the Investment Function

Tier A • Aug ✓ • Map #63 (1/16) • 2 Qs • Vol 1 p.209 • LOs a to c

P1 **a** **Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.**

WORKBOOK A30 + VARDEEZ FOCUS

A30·S2 reserve-management strategy: the net liquidity position as the starting point

P1 **b** **Compare strategies that a bank can use to meet demands for additional liquidity.**

WORKBOOK A30 + VARDEEZ FOCUS

A30·S1 asset liquidity (asset conversion) vs borrowed liquidity (liability management) vs balanced strategy: pros, cons and who uses which, as seen in August

P1 **c** **Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).**

WORKBOOK A26

A26·S2 liquidity requirement built from trend, seasonal and cyclical components, then the liquidity deficit or surplus

LTR-6 Intraday Liquidity Risk Management

P3 ×2 P4 ×1 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *L O seventy-one Explain the uses and sources of intraday liquidity*

RECALL WORKBOOK SECTIONS

D27 **Intraday Liquidity and Dealer Bank Failure Mechanics**

Tier D • Map #65 (1/16) • notes only • Vol 2 p.174 • LOs a to c

P3 **a** Identify and explain the uses and sources of intraday liquidity.

WORKBOOK D27 + VARDEEZ FOCUS

► Vardeez: Uses and sources of intraday liquidity

D27·S2 the payment-system angle: uses and sources of intraday liquidity, daylight overdraft policy, FMU notes defences and tools (notes: LTR-6 bullets)

P4 **b** Discuss the governance structure of intraday liquidity risk management.

P3 **c** Differentiate between methods for tracking intraday flows and monitoring risk levels.

WORKBOOK D27

D27·S1 intraday liquidity monitoring tools: measures of flows versus measures of risk levels (notes: measurement bullets)

LTR-7 Monitoring Liquidity

P1 ×2 P3 ×2 P4 ×1 VARDEEZ: MODERATE WORKBOOK BEST: TIER A

Vardeez: *Deterministic and stochastic cash flows and liquidity option concept. This chapter is you might see one question or you may not see at all from the calculations. So do not get hung up on these calculations - Tsef and tsaa numerical seems imp*

! Maybe one question, maybe none. Do not get hung up on the calculations.

RECALL WORKBOOK SECTIONS

A26 **Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing**

Tier A • Aug ✓ • Map #64 (2/16) • 4 Qs • Vol 1 p.185 • LOs a to e

P3 **a** Differentiate between deterministic and stochastic cash flows and provide examples of each.

VARDEEZ FOCUS

▶ Vardeez: Deterministic vs stochastic cash flows

P3 **b** Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process.

VARDEEZ FOCUS

▶ Vardeez: Liquidity options

P4 c Describe and apply the concepts of liquidity risk, funding cost risk, liquidity generation capacity, expected liquidity, and cash flow at risk.

P1 **d** **Interpret the term structure of expected cash flows and cumulative cash flows.**

WORKBOOK A26 + VARDEEZ FOCUS

▶ Vardeez: Term-structure (TSECF / TSCCF-type) numericals seem important

A26·S1 TSECF and TSAA numerical, plain form of the August recall: which term structures move and by how much

A26·S4 TSECCF with initial cash and the asset sale that keeps the TSLe non-negative (Q2)

P1 **e** **Discuss the impact of available asset transactions on cash flows and liquidity generation capacity.**

WORKBOOK A26

A26·S3 repo of an available asset: cash into the TSCLGC, haircut and accrued interest

A26·S4 TSECCF with initial cash and the asset sale that keeps the TSLe non-negative (Q2)

LTR-8 The Failure Mechanics of Dealer Banks

P3 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *One question might come up from dealer banks. So know it well, maybe go through crash course notes and that would be it*

! One question might come. Crash course notes are enough.

RECALL WORKBOOK SECTIONS

D27 **Intraday Liquidity and Dealer Bank Failure Mechanics**

Tier D • Map #65 (1/16) • notes only • Vol 2 p.174 • LOs a to c

P3 **a** Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business.

VARDEEZ FOCUS

P3 **b** Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks.

WORKBOOK D27 + VARDEEZ FOCUS

D27·S3 dealer-bank runs on repo, on prime-brokerage balances and by OTC derivatives counterparties, ending with the clearing bank (notes: LTR-8 bullets)

P3 **c** Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.

VARDEEZ FOCUS

LTR-9 Liquidity Stress Testing

P3 ×1 P4 ×2 VARDEEZ: MODERATE WORKBOOK BEST: TIER A

Vardeez: *Know the calculation of contingent liquidity asset buffer. Calculation is important And rest just read from crash course notes. You would be good to go*

! Rest: crash course notes.

RECALL WORKBOOK SECTIONS

A26 **Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing**

Tier A • Aug ✓ • Map #64 (2/16) • 4 Qs • Vol 1 p.185 • LOs a to c

P4 a Differentiate between various types of liquidity, including funding, operational, strategic, contingent, and restricted liquidity.

P3 **b** Estimate contingent liquidity via the liquid asset buffer.

VARDEEZ FOCUS

▶ Vardeez: Contingent liquidity via liquid asset buffer: calculation important

P4 c Discuss liquidity stress test design issues such as scope, scenario development, assumptions, outputs, governance, and integration with other risk models.

LTR-10 Liquidity Risk Reporting and Stress Testing

P3 ×1 P4 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Just know the liquidity risk reporting, what are the different reports specifically for liquidity? Just know about it. It's not going to be a detailed. That's going to be tested on the exam. Just know the basics, the interpretation is all that is needed*

RECALL WORKBOOK SECTIONS

A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing

Tier A • Aug ✓ • Map #64 (2/16) • 4 Qs • Vol 1 p.185 • LOs a to c

P4 a Describe best practices for the reporting of a bank's liquidity position.

P3 **b** Compare and interpret different types of liquidity risk reports.

VARDEEZ FOCUS

► Vardeez: Know the different liquidity reports; basics and interpretation only

P4 c Explain the process of reporting a liquidity stress test and interpret a liquidity stress test report.

LTR-11 Contingency Funding Planning

P2 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *contingency funding planning entire chapter is important. Refer the crash course notes and know this chapter*

! Entire chapter important. Refer to crash course notes.

RECALL WORKBOOK SECTIONS

D26 Contingency Funding Planning

Tier D • Map #62 (1/16) • notes only • Vol 2 p.171 • LOs a to c

P2 **a** **Discuss the relationship between contingency funding planning and liquidity stress testing.**

VARDEEZ FOCUS

P2 **b** **Describe best practices in the design of a sound contingency funding plan.**

WORKBOOK D26 + VARDEEZ FOCUS

D26·S4 testing requirements and operational readiness (notes: testing)

P2 **c** **Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting).**

WORKBOOK D26 + VARDEEZ FOCUS

D26·S1 governance of the CFP: who designs, invokes, oversees and approves (notes: governance and roles)

D26·S2 trigger design and escalation: EWIs, liquidity health measures, escalation levels (notes: monitoring notes and escalation)

D26·S3 the menu of contingent actions and their reliability under stress, including signalling effects (notes: notes contingent actions, market signals)

LTR-12 Managing Non-Deposit Liabilities

P1 ×3 P3 ×1 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Calculating available funds gap and calculating overall cost of funds using historical average and pooled funds approach*

RECALL WORKBOOK SECTIONS

A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing

Tier A • Aug ✓ • Map #47 (2/16) • TG ✓ • 3 Qs • Vol 1 p.192 • LOs a to d

P1 **a** **Distinguish between the various sources of non-deposit liabilities at a bank.**

WORKBOOK A27

A27·S2 sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August

P3 **b** Describe and calculate the available funds gap.

VARDEEZ FOCUS

▶ Vardeez: Available funds gap calculation

P1 **c** **Discuss factors affecting the choice of non-deposit funding sources.**

WORKBOOK A27

A27·S2 sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August

P1 **d** **Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.**

WORKBOOK A27 + VARDEEZ FOCUS

▶ Vardeez: Cost of funds: historical average and pooled-funds approaches

A27·S1 pooled-funds approach (forward looking hurdle rate) against the historical average cost approach, numerical

LTR-13 Repurchase Agreements and Financing

P2 ×2 P3 ×2 P4 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *For this, understand general collateral, special collateral and LO78F. Special spreads and the auction cycle*

RECALL WORKBOOK SECTIONS

D6 **Repurchase Agreements**

Tier D • Map #43 (2/16) • 2 Qs • Vol 2 p.101 • LOs a to g

P3 **a** Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction.

WORKBOOK D6

D6·S1 repo settlement amount from the closed-book template, with accrued interest and haircut, and the repurchase price at the unwind

P4 b Discuss common motivations for entering into repos, including their use in cash management and liquidity management.

P3 **c** Explain how counterparty risk and liquidity risk can arise through the use of repo transactions.

WORKBOOK D6

D6·S2 which party is more vulnerable when a bank will not roll the repo over (Q2)

P4 d Assess the role of repo transactions in the collapses of Lehman Brothers and Bear Stearns during the 2007-2009 financial crisis.

P2 **e** **Compare the use of general and special collateral in repo transactions.**

VARDEEZ FOCUS

▶ Vardeez: General vs special collateral

P2 **f** **Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle.**

VARDEEZ FOCUS

▶ Vardeez: Special spreads and the auction cycle (his ”LO 78.f”)

P4 g Calculate the financing advantage of a bond trading special when used in a repo transaction.

LTR-14 Liquidity Transfer Pricing: A Guide to Better Practice

P1 ×2 P4 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Understand the process of liquidity transfer pricing and know the calculation of marginal cost of funds approach. Important*

RECALL WORKBOOK SECTIONS

A27 **Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing**

Tier A • Aug ✓ • Map #47 (2/16) • TG ✓ • 3 Qs • Vol 1 p.192 • LOs a to d

P1	a	**Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.** WORKBOOK A27 + VARDEEZ FOCUS ▶ Vardeez: Process of LTP **A27·S4** LTP's role in charging liquidity costs back to business lines, including the liquidity cushion (Q3)
P4	b	Discuss challenges that may arise for banks during the implementation of LTP.
P1	c	**Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost).** WORKBOOK A27 + VARDEEZ FOCUS ▶ Vardeez: Marginal cost of funds approach: calculation **A27·S3** pooled average cost of funds against matched-maturity marginal cost in LTP
P4	d	Describe the contingent liquidity risk pricing process and calculate the cost of contingent liquidity risk.

LTR-15 The US Dollar Shortage in Global Banking and the International Policy Response

P2 ×3 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *Short Reading and anything can be tested here - so know it well*

! Short reading; anything can be tested.

RECALL WORKBOOK SECTIONS

D8 **US Dollar Shortage and the Cross-Currency Basis**

Tier D • Map #46 (2/16) • TG ✓ • 2 Qs • Vol 2 p.110 • LOs a to c

P2	a	**Identify the causes of the U.S. dollar shortage during the Great Financial Crisis.** WORKBOOK D8 + VARDEEZ FOCUS D8·S1 why non-US (chiefly European) banks faced a US dollar shortage in the GFC
P2	b	**Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities.** VARDEEZ FOCUS
P2	c	**Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness.** VARDEEZ FOCUS

LTR-16 Covered Interest Parity Lost: Understanding the Cross-Currency Basis

P3 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *Short Reading and anything can be tested here - so know it well*

! Short reading; anything can be tested.

RECALL WORKBOOK SECTIONS

D8 **US Dollar Shortage and the Cross-Currency Basis**

Tier D • Map #46 (2/16) • TG ✓ • 2 Qs • Vol 2 p.110 • LOs a to c

P3	a	Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. VARDEEZ FOCUS
P3	b	Identify key factors that affect the cross-currency swap basis. WORKBOOK D8 + VARDEEZ FOCUS D8·S2 covered interest parity violations and the cross-currency basis, the adjacent reading (Q2)
P3	c	Assess the causes of covered interest rate parity violations after the financial crisis of 2008. WORKBOOK D8 + VARDEEZ FOCUS D8·S2 covered interest parity violations and the cross-currency basis, the adjacent reading (Q2)

LTR-17 Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

P1 ×2 P2 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Entire Chapter and Each Learning Objective is Important - so Do not Ignore this Chapter*

! Entire chapter and every LO important. Do not ignore.

RECALL WORKBOOK SECTIONS

A16 Duration Gap and Interest-Sensitive Gap

Tier A • Aug ✓ • Map #24 (5/16) • TG ✓ • 4 Qs • Vol 1 p.121 • LOs a to d

P2 **a** **Discuss how asset-liability management strategies can help a bank hedge against interest rate risk.**

VARDEEZ FOCUS

P1 **b** **Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.**

WORKBOOK A16 + VARDEEZ FOCUS

A16·S4 IS gap and ISR: cumulative gap and the change in net interest income

P1 **c** **Describe duration gap management and apply this strategy to protect a bank's net worth.**

WORKBOOK A16 + VARDEEZ FOCUS

A16·S1 duration gap, plain form of the August recall

A16·S2 impact on net worth given asset and liability durations, with direction

A16·S3 duration of assets and liabilities: the liability duration that immunises net worth

A16·S5 polarity of the leverage-adjusted gap against a rate move, including a sign reversal from the leverage adjustment (Q1, Q2)

P2 **d** **Discuss the limitations of interest-sensitive gap management and duration gap management.**

VARDEEZ FOCUS

Risk Management and Investment Management

15% • 12 Qs

IM-1 Factor Theory

P3 ×3 P4 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER B

Vardeez: *CAPM assumptions and limitations; CAPM vs multifactor models*

RECALL WORKBOOK SECTIONS

B4 Factor Theory, Fama-French and Style

Tier B • Map #18 (5/16) • TG ✓ • 2 Qs • Vol 2 p.29 • LOs a to f

P4 a Describe factors that impact asset prices and explain the theory of factor risk premiums.

P3 **b** Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM.

VARDEEZ FOCUS

► Vardeez: CAPM assumptions and limitations

P3 **c** Explain the implications of using the CAPM to value assets, including equilibrium and optimal holdings, exposure to factor risk, its treatment of diversification benefits, and shortcomings of the CAPM.

VARDEEZ FOCUS

► Vardeez: CAPM limitations

P3 **d** Describe multifactor models and compare and contrast multifactor models to the CAPM.

VARDEEZ FOCUS

► Vardeez: CAPM vs multifactor models

P4 e Explain how stochastic discount factors are created and apply them in the valuation of assets.

P4 f Describe efficient market theory and explain how markets can be inefficient.

IM-2 Factors

P2 ×2 P3 ×1 P4 ×2 VARDEEZ: UNTIERED WORKBOOK BEST: TIER B

Vardeez: *LO 84.C is Imp and LO 84. E is imp which is value vs momentum strategies,*

RECALL WORKBOOK SECTIONS

B4 Factor Theory, Fama-French and Style

Tier B • Map #18 (5/16) • TG ✓ • 2 Qs • Vol 2 p.29 • LOs a to e

D18 Inflation and Market Stress Across Bond Maturities

Tier D • Map #72 (2/16) • TG ✓ • 1 Qs • Vol 2 p.146 • LOs b

P4 a Describe the process of value investing and explain why a value premium may exist.

P4 b Explain how different macroeconomic risk factors, including economic growth, inflation, and volatility, affect asset returns and risk premiums.

P3 **c** Assess methods of mitigating volatility risk in a portfolio and describe challenges that arise when managing volatility risk.

VARDEEZ FOCUS

► Vardeez: (his ”LO 84.c”)

P2 **d** **Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example.**

WORKBOOK B4

B4·S1 SMB (small minus big) vs SML (security market line), with HML, in a Fama-French portfolio composition

B4·S2 investing style (style, investment or dynamic factors) as a theory item

P2 **e** **Compare value and momentum investment strategies, including their return and risk profiles.**

WORKBOOK B4 + VARDEEZ FOCUS

► Vardeez: Value vs momentum strategies

B4·S3 value vs momentum: return and risk profiles

IM-3 Alpha (and the Low-Risk Anomaly)

P1 ×2 P2 ×5 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *1. Grinold's Fundamental Law (IC, Breadth, IR) 2) Calculation of FAMA French Model using SMB,HML (Pls know this very well) and 3) Style Analysis*

RECALL WORKBOOK SECTIONS

A18 Performance Measures and Attribution

Tier A • Aug ✓ • Map #36 (5/16) • TG ✓ • 6 Qs • Vol 1 p.132 • LOs d

B4 Factor Theory, Fama-French and Style

Tier B • Map #18 (5/16) • TG ✓ • 2 Qs • Vol 2 p.29 • LOs i

P2	a	**Describe and evaluate the low-risk anomaly of asset returns.** WORKBOOK B4 B4·S4 the low-risk anomaly: volatility anomaly vs beta anomaly and their explanations (Q2)
P1	b	**Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio.** WORKBOOK A18, C4 A18·S3 Jensen's alpha and the information ratio C4·S4 IR and active risk numerical: IR from expected excess return and TEV, portfolio IR
P4	c	Explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha.
P1	d	**Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law.** WORKBOOK A18 + VARDEEZ FOCUS ▶ Vardeez: Grinold's fundamental law (IC, breadth, IR) A18·S5 Grinold's fundamental law and the maximum attainable information ratio
P2	e	**Apply a factor regression to construct a benchmark with multiple factors, measure a portfolio's sensitivity to those factors, and measure alpha against that benchmark.** WORKBOOK B4 + VARDEEZ FOCUS ▶ Vardeez: Fama-French with SMB and HML: know very well B4·S1 SMB (small minus big) vs SML (security market line), with HML, in a Fama-French portfolio composition
P2	f	**Explain how to use style analysis to handle time-varying factor exposures.** VARDEEZ FOCUS ▶ Vardeez: Style analysis
P4	g	Describe issues that arise when measuring alphas for nonlinear strategies.
P2	h	**Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly.** WORKBOOK B4 B4·S4 the low-risk anomaly: volatility anomaly vs beta anomaly and their explanations (Q2)
P2	i	**Describe potential explanations for the risk anomaly.** WORKBOOK B4 B4·S4 the low-risk anomaly: volatility anomaly vs beta anomaly and their explanations (Q2)

IM-4 Portfolio Construction

P2 ×3 P3 ×2 P4 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

Vardeez: *1. Transaction Costs and No-Trade Region 2. Portfolio Construction Techniques*

RECALL WORKBOOK SECTIONS

D28 Portfolio Construction and the No-Trade Region

Tier D • Map #66 (1/16) • notes only • Vol 2 p.178 • LOs a to i

P4 a Describe the inputs to the portfolio construction process and explain challenges faced when using these inputs.

P3 **b** Evaluate the motivation for and the methods used for refining alphas in the implementation process.

WORKBOOK D28

D28·S1 alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry)

P3 **c** Describe neutralization and the different approaches used for refining alphas to be neutral.

WORKBOOK D28

D28·S1 alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry)

P2 **d** **Explain the implications of transaction costs on portfolio construction.**

WORKBOOK D28 + VARDEEZ FOCUS

▶ Vardeez: Transaction costs

D28·S2 transaction costs: amortisation over the holding period

P4 e Describe practical issues in portfolio construction, including the determination of an appropriate risk aversion, aversions to specific risks, and proper alpha coverage.

P4 f Describe portfolio revisions and rebalancing, and analyze the tradeoffs between alpha, risk, transaction costs, and time horizon.

P2 **g** **Determine the optimal no-trade region for rebalancing with transaction costs.**

WORKBOOK D28 + VARDEEZ FOCUS

▶ Vardeez: No-trade region

D28·S3 the optimal no-trade region for MCVA and for alpha, and Leland's band around a target allocation

P2 **h** **Evaluate the strengths and weaknesses of the following portfolio construction techniques: screens, stratification, linear programming, and quadratic programming.**

WORKBOOK D28 + VARDEEZ FOCUS

▶ Vardeez: Portfolio construction techniques

D28·S4 the four techniques as a four-statement audit: screens, stratification, LP, QP

P4 i Describe dispersion, explain its causes, and describe methods for controlling forms of dispersion.

IM-5 Portfolio Risk: Analytical Methods

P1 ×3 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *1. Marginal, Incremental and Component VaR Calculations 2. Diversified vs Undiversified VaR*

RECALL WORKBOOK SECTIONS

A12 Incremental, Marginal and Component VaR

Tier A • Aug ✓ • Map #13 (6/16) • TG ✓ • 4 Qs • Vol 1 p.94 • LOs a to e

P1 **a** **Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR.**

WORKBOOK A12 + VARDEEZ FOCUS

▶ Vardeez: Marginal, incremental and component VaR calculations; diversified vs undiversified VaR

A12·S1 incremental VaR of including a platinum position in a silver book, full revaluation versus the marginal VaR shortcut

A12·S3 marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio

A12·S4 closed-book formulas: marginal VaR = (VaR/W)β; component VaR = individual VaR × ρ

A12·S6 intermediate-result trap between marginal and component VaR (Q3; tangent versus curve in Q1)

P4 b Explain the impact of correlation on portfolio risk.

P1 **c** **Apply the concept of marginal VaR in making portfolio management decisions.**

WORKBOOK A12, A18

A12·S2 allocation using marginal VaR: where to add (Q2) and which position to grow for a better return-torisk trade-off

A12·S5 shortcut: when ranking marginal VaRs only beta matters

A18·S2 marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio (A12.4, see also)

P1 **d** **Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk.**

WORKBOOK A12

A12·S2 allocation using marginal VaR: where to add (Q2) and which position to grow for a better return-torisk trade-off

P4 e Explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management.

IM-6 VaR and Risk Budgeting in Investment Management

P2 ×2 P4 ×6 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *Suplus at Risk Calculation and LO 88.h - Risk Budegeting Calucaltions*

RECALL WORKBOOK SECTIONS

C4 Policy-Mix VaR and Risk Budgeting

Tier C • Map #35 (4/16) • TG ✓ • 3 Qs • Vol 2 p.50 • LOs a to h

P4	a	Define risk budgeting.
P4	b	Describe the impact of horizon, turnover, and leverage on the risk management process in the investment management industry.
P4	c	Describe the investment process of large investors such as pension funds.
P4	d	Describe the risk management challenges associated with investments in hedge funds.
P2	e	**Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk.** WORKBOOK C4 + VARDEEZ FOCUS ▶ Vardeez: Surplus at risk calculation C4·S1 VaR calculated by the policy-mix method: policy weights on benchmark returns, diversified C4·S2 the six-way sibling set: absolute, relative, policy-mix, active-management, funding, sponsor risk
P4	f	Explain the use of VaR to monitor risk.
P4	g	Explain how VaR can be used in the development of investment guidelines and for improving the investment process.
P2	h	**Describe the risk budgeting process and calculate risk budgets across asset classes and active managers.** WORKBOOK C4 + VARDEEZ FOCUS ▶ Vardeez: Risk budgeting calculations (his "LO 88.h") C4·S3 portfolio return computation: expected excess return of a pool of managers C4·S4 IR and active risk numerical: IR from expected excess return and TEV, portfolio IR C4·S5 risk budgets across asset classes and across active managers

IM-7 Portfolio Performance Evaluation

P1 ×3 P2 ×2 P4 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *ime-Weighted vs Dollar-Weighted Returns, M² Calculation and Style Analysis which is Last LO in this chapter*

! Vardeez calls style analysis "the last LO". In the 2026 official order it is IM-7 e, not i.

RECALL WORKBOOK SECTIONS

A18 Performance Measures and Attribution

Tier A • Aug ✓ • Map #36 (5/16) • TG ✓ • 6 Qs • Vol 1 p.132 • LOs a to i

P2	a	**Differentiate between the time-weighted and dollar-weighted returns of a portfolio and describe their appropriate uses.** VARDEEZ FOCUS ▶ Vardeez: Time-weighted vs dollar-weighted returns
P1	b	**Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant.** WORKBOOK A18 **A18·S1** Sharpe ratio, the harder form: Sharpe paired with M² (Q4); Sharpe as the wrong lens for one of many subportfolios **A18·S2** marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio (A12.4, see also) **A18·S3** Jensen's alpha and the information ratio **A18·S6** which measure suits which use: Sharpe, Treynor, information ratio (Q2, Q3, Q6)
P1	c	**Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures.** WORKBOOK A18 + VARDEEZ FOCUS ▶ Vardeez: M-squared calculation **A18·S1** Sharpe ratio, the harder form: Sharpe paired with M² (Q4); Sharpe as the wrong lens for one of many subportfolios **A18·S6** which measure suits which use: Sharpe, Treynor, information ratio (Q2, Q3, Q6)
P4	d	Determine the statistical significance of a performance measure using standard error and the t-statistic.
P2	e	**Describe style analysis.** VARDEEZ FOCUS ▶ Vardeez: Style analysis
P4	f	Explain the difficulties in measuring the performance of actively managed portfolios.
P4	g	Describe performance manipulation and the problems associated with using conventional performance measures.
P4	h	Describe techniques to measure the market timing ability of fund managers with a regression and with a call option model and calculate a manager's return due to market timing.
P1	i	**Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution.** WORKBOOK A18 **A18·S4** attribution split into asset allocation and security selection effects

IM-8 Hedge Fund Investment Strategies

P1 ×4 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *1. Convertible Arbitrage 2. Merger Arbitrage 3. Equity-Based and Macro Hedge Fund Strategies*

RECALL WORKBOOK SECTIONS

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff

Tier A • Aug ✓ • Map #34 (3/16) • 4 Qs • Vol 1 p.172 • LOs a to d

P1 **a** **Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Equity-based and macro strategies

A24·S1 hedge fund strategies: equity, macro, arbitrage, event-driven, distressed

P1 **b** **Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Convertible arbitrage

A24·S1 hedge fund strategies: equity, macro, arbitrage, event-driven, distressed

P1 **c** **Describe the mechanics and risks of event-driven strategies such as merger arbitrage.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Merger arbitrage

A24·S1 hedge fund strategies: equity, macro, arbitrage, event-driven, distressed

P1 **d** **Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy.**

WORKBOOK A24

A24·S1 hedge fund strategies: equity, macro, arbitrage, event-driven, distressed

IM-9 Risk, Regulation and Organizational Structure

P1 ×4 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *1. Hedge Fund Risks and Systemic Risk 2. Hedge Fund Regulatory Approaches*

RECALL WORKBOOK SECTIONS

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff

Tier A • Aug ✓ • Map #34 (3/16) • 4 Qs • Vol 1 p.172 • LOs a to d

P1 **a** **Describe different risks faced by investors in hedge funds.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Hedge fund risks

A24·S2 hedge fund risks, systemic risk, regulation and structure

P1 **b** **Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Systemic risk

A24·S2 hedge fund risks, systemic risk, regulation and structure

P1 **c** **Summarize different approaches taken by regulators to mitigate risks related to hedge funds.**

WORKBOOK A24 + VARDEEZ FOCUS

▶ Vardeez: Regulatory approaches

A24·S2 hedge fund risks, systemic risk, regulation and structure

P1 **d** **Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure.**

WORKBOOK A24

A24·S2 hedge fund risks, systemic risk, regulation and structure

IM-10 The Rise and Risks of Private Credit

P1 ×2 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Private Credit Risks*

RECALL WORKBOOK SECTIONS

A28 Private Credit, Private Markets and Their Drivers

Tier A • Aug ✓ • Map #49 (2/16) • TG ✓ • 4 Qs • Vol 1 p.198 • LOs a to d

P4 a Describe characteristics of private credit, including its typical investors and borrowers, and compare private credit to other types of loans and fixed-income instruments.

P4 b Explain the return profile and growth profile of the private credit asset class, and compare the historical returns of private credit to those of other asset classes.

P1 **c Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability.**

WORKBOOK A28 + VARDEEZ FOCUS

▶ Vardeez: Private credit risks

A28·S3 valuation opacity: stale marks, deferred losses, NAV-based metrics

A28·S4 financial-stability concerns and the IMF's policy fixes

P1 **d Assess potential policy recommendations that could help mitigate the risks associated with private credit.**

WORKBOOK A28

A28·S4 financial-stability concerns and the IMF's policy fixes

IM-11 Private Markets Investing

P1 ×4 P4 ×2 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Private Market Investment Structures, Private Equity Performance Measures: IRR, MOIC, PME, Co-Investing: Benefits and Risks*

RECALL WORKBOOK SECTIONS

A28 Private Credit, Private Markets and Their Drivers

Tier A • Aug ✓ • Map #49 (2/16) • TG ✓ • 4 Qs • Vol 1 p.198 • LOs a to f

P4 a Explain the motivations for investors to invest in private markets.

P1 **b Describe different investment structures that can be used to invest in private assets and compare their characteristics.**

WORKBOOK A28 + VARDEEZ FOCUS

▶ Vardeez: Investment structures

A28·S6 private markets structures, co-investing and intermediated vehicles

P1 **c Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages.**

WORKBOOK A28 + VARDEEZ FOCUS

▶ Vardeez: IRR, MOIC, PME

A28·S5 private equity performance measures: IRR vs TVPI and DPI vs PME

P4 d Explain how different risks can arise for investors in private markets and assess how these risks can be measured.

P1 **e Describe the practice of co-investing in private markets and assess its advantages and disadvantages.**

WORKBOOK A28 + VARDEEZ FOCUS

▶ Vardeez: Co-investing: benefits and risks

A28·S6 private markets structures, co-investing and intermediated vehicles

P1 **f Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund.**

WORKBOOK A28

A28·S6 private markets structures, co-investing and intermediated vehicles

IM-12 Performing Due Diligence on Specific Managers and Funds

P3 ×2 P4 ×5 VARDEEZ: MODERATE WORKBOOK BEST: TIER A

Vardeez: *LO 94.c - Manager Evaluation, and LO 94.e - Operational Due Diligence*

RECALL WORKBOOK SECTIONS

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff

Tier A • Aug ✓ • Map #34 (3/16) • 4 Qs • Vol 1 p.172 • LOs a to g

P4 a Identify reasons for the failures of hedge funds in the past.

P4 b Explain elements of the due diligence process used to assess investment managers.

P3 **c** Identify themes and questions investors can consider when evaluating a hedge fund manager.

VARDEEZ FOCUS

▶ Vardeez: Manager evaluation

P4 d Describe criteria that can be evaluated in assessing a hedge fund's risk management process.

P3 **e** Explain how due diligence can be performed on a hedge fund's operational environment.

VARDEEZ FOCUS

▶ Vardeez: Operational due diligence

P4 f Explain how a hedge fund's business model risk and its fraud risk can be assessed.

P4 g Describe elements that can be included as part of a due diligence questionnaire.

IM-13 Private Capital Investing: Distress Symptoms and Remedies

P3 ×2 P4 ×1 VARDEEZ: UNTIERED NO WORKBOOK SECTION

Vardeez: *Lo 95.a and LO 95.b both are Imp*

RECALL WORKBOOK SECTIONS

A28 Private Credit, Private Markets and Their Drivers

Tier A • Aug ✓ • Map #49 (2/16) • TG ✓ • 4 Qs • Vol 1 p.198 • cross-reference

P3 **a** Explain different reasons why firms may experience financial distress and identify potential indicators of distress.

VARDEEZ FOCUS

P3 **b** Describe methods firms can use to remedy a financial distress situation at different stages of distress and explain the potential implications and challenges to using each method.

VARDEEZ FOCUS

P4 c Summarize potential legal challenges that can arise when a firm is in financial distress or is going through a restructuring.

IM-14 Madoff: A Riot of Red Flags

P1 ×1 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Know this Well and expect 1 question from this*

! Know this well; expect one question.

RECALL WORKBOOK SECTIONS

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff

Tier A • Aug ✓ • Map #34 (3/16) • 4 Qs • Vol 1 p.172 • LOs a, b

P2 **a** **Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance.**

VARDEEZ FOCUS

P1 **b** **Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk.**

WORKBOOK A24 + VARDEEZ FOCUS

A24·S4 Madoff red flags as a case↔lesson matching item against other cases (Q4)

IM-15 Market-Driven Scenarios: An Approach for Plausible Scenario Construction

P2 ×2 P3 ×1 P4 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER C

Vardeez: *Mahalanobis Distance, Scenario, Volatility and Correlation Z-Scores (remember Learning Objective is EXPLAIN - so need to get hung up on Formulas), Market-Driven Scenario Construction (know how to create it)*

RECALL WORKBOOK SECTIONS

C9 Stress-Testing Practices and Scenario Design

Tier C • Map #38 (3/16) • TG ✓ • 2 Qs • Vol 2 p.74 • LOs a to d

P3 **a** Describe and evaluate the market-driven scenario (MDS) approach used to develop scenarios for stress testing investment portfolios.

WORKBOOK C9

C9·S2 market-driven scenarios: plausible scenario construction, z-scores and covariance choice

P2 **b** **Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios.**

WORKBOOK C9 + VARDEEZ FOCUS

▶ Vardeez: Mahalanobis distance, scenario / volatility / correlation z-scores. LO verb is EXPLAIN: no need to get hung up on formulas

C9·S2 market-driven scenarios: plausible scenario construction, z-scores and covariance choice

P2 **c** **Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided.**

WORKBOOK C9 + VARDEEZ FOCUS

▶ Vardeez: Know how to construct a market-driven scenario

C9·S2 market-driven scenarios: plausible scenario construction, z-scores and covariance choice

P4 d Describe challenges and considerations that may arise when implementing the market-driven scenario approach.

IM-16 Liquidity Risk Management

P2 ×5 VARDEEZ: VERY IMPORTANT NO WORKBOOK SECTION

Vardeez: *LO 98.b is imp and rest all Learning Objectives - pls know it*

! His "LO 98.b" is flagged as the important one; know the rest too.

P2 **a** **Explain best practices for managing liquidity risk in an investment fund.**

VARDEEZ FOCUS

P2 **b** **Explain potential challenges fund managers face when managing liquidity risk in fixed-income portfolios, including challenges related to data modeling.**

VARDEEZ FOCUS

▶ Vardeez: Flagged as the key LO

P2 **c** **Describe and apply different approaches to model asset liquidity, including an approach to modeling infrequently traded bonds and the t-cost model to calculate transaction costs for corporate bonds.**

VARDEEZ FOCUS

P2 **d** **Identify examples of extraordinary measures that regulators may allow funds to take to meet unanticipated redemption requests, and explain when each measure may be allowed.**

VARDEEZ FOCUS

P2 **e** **Explain approaches to manage and model redemption risk in an investment fund, including the redemption waterfall, approaches to model a fund's level of redemption-at-risk, and approaches to optimize liquidity to meet redemption requests.**

VARDEEZ FOCUS

IM-17 Illiquid Assets

P1 ×1 P2 ×2 P3 ×1 P4 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *LO 99.e and LO 99.f is Important*

RECALL WORKBOOK SECTIONS

A24 **Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff**

Tier A • Aug ✓ • Map #34 (3/16) • 4 Qs • Vol 1 p.172 • LOs c

D9 **Illiquid Assets**

Tier D • Map #48 (2/16) • 2 Qs • Vol 2 p.115 • LOs a to f

P4	a	Evaluate the characteristics of illiquid markets.
P4	b	Discuss the relationship between market imperfections and illiquidity.
P1	c	**Assess the impact of biases on reported returns for illiquid assets.** WORKBOOK A24, D9 **A24·S3** survivorship bias against its siblings: selection, reporting, backfill D9·S1 biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing
P3	d	Explain the process of unsmoothing returns and the effects of unsmoothing. WORKBOOK D9 D9·S1 biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing
P2	e	**Compare illiquidity risk premiums across and within asset categories.** WORKBOOK D9 + VARDEEZ FOCUS D9·S2 illiquidity risk premiums across versus within asset classes, and the four ways to harvest them
P2	f	**Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets.** WORKBOOK D9 + VARDEEZ FOCUS D9·S3 portfolio allocation with illiquid assets: optimal holdings, hurdle rates, rebalancing point

Current Issues in Financial Markets

10% • 8 Qs

Vardeez (all CI readings): *Please Solve Book Wise Current Issues Questions as anything can show up*

CI-1 Advances in Artificial Intelligence: Implications for Capital Markets Activities

P1 ×2 P2 ×2 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

Vardeez: *Please Solve Book Wise Current Issues Questions as anything can show up*

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

A14 Artificial Intelligence and GenAI

Tier A • Aug ✓ • Map #20 (6/16) • TG ✓ • 3 Qs • Vol 1 p.107 • LOs a to d

P1 **a** **Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI.**

WORKBOOK A14 + VARDEEZ FOCUS

A14·S1 AI vs GenAI: the IMF's two model groups, what GenAI adds, and its signature failure mode

A14·S3 application of AI in market risk: volatility and liquidity models, risk scenarios, dynamic risk sensing

P2 **b** **Explain the implications of further adoption of AI on market dynamics.**

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P1 **c** **Explain the implications of further adoption of AI on financial stability.**

WORKBOOK A14 + VARDEEZ FOCUS

A14·S2 disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors

A14·S4 financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk

P2 **d** **Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities.**

VARDEEZ FOCUS

CI-2 The Financial Stability Implications of Artificial Intelligence

P1 ×2 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

A14 Artificial Intelligence and GenAI

Tier A • Aug ✓ • Map #20 (6/16) • TG ✓ • 3 Qs • Vol 1 p.107 • LOs a to c

P1 **a** **Explain key developments in AI since 2017, differentiating between supply-side drivers and demand-side drivers.**

WORKBOOK A14 + VARDEEZ FOCUS

A14·S1 AI vs GenAI: the IMF's two model groups, what GenAI adds, and its signature failure mode

P2 **b** **Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases.**

VARDEEZ FOCUS

P1 **c** **Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: - Third-party dependencies and service provider concentration - Market correlations - Cyber risks - Model risk, data, and governance - Other vulnerabilities, such as fraud, disinformation, and misalignment**

WORKBOOK A14 + VARDEEZ FOCUS

A14·S2 disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors

A14·S4 financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk

CI-3 The Global Drivers of Private Credit

P1 ×2 P2 ×1 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER A

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

A28 Private Credit, Private Markets and Their Drivers

Tier A • Aug ✓ • Map #49 (2/16) • TG ✓ • 4 Qs • Vol 1 p.198 • LOs a to c

P2 **a** **Describe the structure and characteristics of private credit funds, as well as the patterns in the global growth of the asset class.**

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P1 **b** **Discuss the potential drivers of private credit growth and interpret regression results about their impact.**

WORKBOOK A28 + VARDEEZ FOCUS

A28·S1 growth drivers of private credit and the BIS regression results

P1 **c** **Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them.**

WORKBOOK A28 + VARDEEZ FOCUS

A28·S2 bank disintermediation: private credit vs bank loans, cost of equity, cost of debt and leverage

CI-4 Global Financial Stability Report

P3 ×6 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

D30 Geopolitical Risk and Financial Stability

Tier D • 2 Qs • Vol 2 p.184 • LOs a to f

P3 **a** Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences.

WORKBOOK D30 + VARDEEZ FOCUS

D30·S1 channels from geopolitical risk to asset prices and financial stability, and policy responses

P3 **b** Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters.

WORKBOOK D30 + VARDEEZ FOCUS

D30·S4 sovereign risk premiums, and commodity importers vs exporters (Q2)

P3 **c** Assess the stock market responses to different domestic and global geopolitical risk events, including Russia's invasion of Ukraine and the U.S.-China trade tensions, and explain the cross-border effects of these events.

VARDEEZ FOCUS

P3 **d** Evaluate the effects of geopolitical shocks on sovereign risk premiums.

WORKBOOK D30 + VARDEEZ FOCUS

D30·S4 sovereign risk premiums, and commodity importers vs exporters (Q2)

P3 **e** Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail- risk premiums.

WORKBOOK D30 + VARDEEZ FOCUS

D30·S3 pricing in equity and option markets (GPR betas, downside and tail-risk premiums)

P3 **f** Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows.

WORKBOOK D30 + VARDEEZ FOCUS

D30·S2 effects on banks and nonbank financial institutions: valuations, lending, fund flows

CI-5 Monetary and fiscal policy: safeguarding stability and trust

P3 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

D16 Monetary and Fiscal Policy

Tier D • Map #67 (2/16) • TG ✓ • 1 Qs • Vol 2 p.140 • LOs a to c

P3 **a** Compare and contrast the channels through which fiscal policy and monetary policy influence a country's economic activity and financial markets, and define the "region of stability" in terms of their joint policy stances.

WORKBOOK D16 + VARDEEZ FOCUS

D16·S1 fiscal and monetary policy: how the two interact, including the tension that high debt creates (fiscal dominance)

P3 **b** Describe the consequences of breaching the boundaries of the region of stability, and how these consequences have evolved over time in advanced economies and in emerging market economies.

VARDEEZ FOCUS

P3 **c** Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy.

WORKBOOK D16 + VARDEEZ FOCUS

D16·S2 LO 104.c: the interest-growth differential and the risks of high public debt (Q1)

CI-6 Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets

P3 ×4 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

D29 Crypto, Tokenisation and Digital Resilience

Tier D • Map #68 (1/16) • notes only • Vol 2 p.181 • LOs a to d

P3 **a** Define and describe crypto assets, including the categories broadly used by global financial regulators to classify them.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S1 CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS notes treatment, Bali Fintech Agenda

P3 **b** Evaluate the key components within the crypto ecosystem, the potential risks generated by these components, and potential regulatory responses to address those risks.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S1 CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS notes treatment, Bali Fintech Agenda

P3 **c** Identify and describe some of the global approaches to the regulation of unbacked crypto assets, including the BCBS' proposed treatment of banks' exposures to crypto assets.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S1 CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS notes treatment, Bali Fintech Agenda

P3 **d** Examine the considerations and steps introduced by the Bali Fintech Agenda (BFA) for developing a regulatory framework for crypto assets.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S1 CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS notes treatment, Bali Fintech Agenda

CI-7 Tokenization and Financial Market Inefficiencies

P2 ×5 VARDEEZ: VERY IMPORTANT WORKBOOK BEST: TIER D

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

D29 Crypto, Tokenisation and Digital Resilience

Tier D • Map #68 (1/16) • notes only • Vol 2 p.181 • LOs a to e

P2 **a** **Explain the process of tokenization and describe the fundamental features of tokenized assets and digital ledgers.**

WORKBOOK D29 + VARDEEZ FOCUS

D29·S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and notes broker effects, ledger models and trust failures

P2 **b** **Describe frictions and inefficiencies that can arise during different phases of an asset's lifecycle and explain how the different features of tokenization can both increase and mitigate risks related to these frictions.**

WORKBOOK D29 + VARDEEZ FOCUS

D29·S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and notes broker effects, ledger models and trust failures

P2 **c** **Explain how the use of tokenization and digital ledgers can impact financial market externalities, including those related to the transmission of shocks across the financial system, investments in market infrastructure, and network and knowledge effects.**

WORKBOOK D29 + VARDEEZ FOCUS

D29·S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and notes broker effects, ledger models and trust failures

P2 **d** **Explain how the proliferation of distributed ledger technologies can impact retail investors and brokerage firms.**

WORKBOOK D29 + VARDEEZ FOCUS

D29·S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and notes broker effects, ledger models and trust failures

P2 **e** **Compare different models and methods that are currently available to achieve sharedness, programmability, and trust on a ledger, and describe the conditions under which trust can fail in such models.**

WORKBOOK D29 + VARDEEZ FOCUS

D29·S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and notes broker effects, ledger models and trust failures

CI-8 Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector

P3 ×3 VARDEEZ: IMPORTANT WORKBOOK BEST: TIER D

! Solve book-wise Current Issues questions; anything can show up.

RECALL WORKBOOK SECTIONS

D29 Crypto, Tokenisation and Digital Resilience

Tier D • Map #68 (1/16) • notes only • Vol 2 p.181 • LOs a to c

P3 **a** Describe characteristics of cyber risks and information/communication technology (ICT) risks faced by financial institutions.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S3 CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools notes and their challenges

P3 **b** Assess the interactions between cyber and IC T risks and financial risks and explain how cyber and IC T risk events at financial institutions can lead to systemic financial risk.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S3 CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools notes and their challenges

P3 **c** Describe potential macroprudential tools and policy measures that can be used to address cyber risks and ICT risks and explain challenges to the adoption of each one.

WORKBOOK D29 + VARDEEZ FOCUS

D29·S3 CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools notes and their challenges